MODULE 3

VARIABILITY

OUTLINE

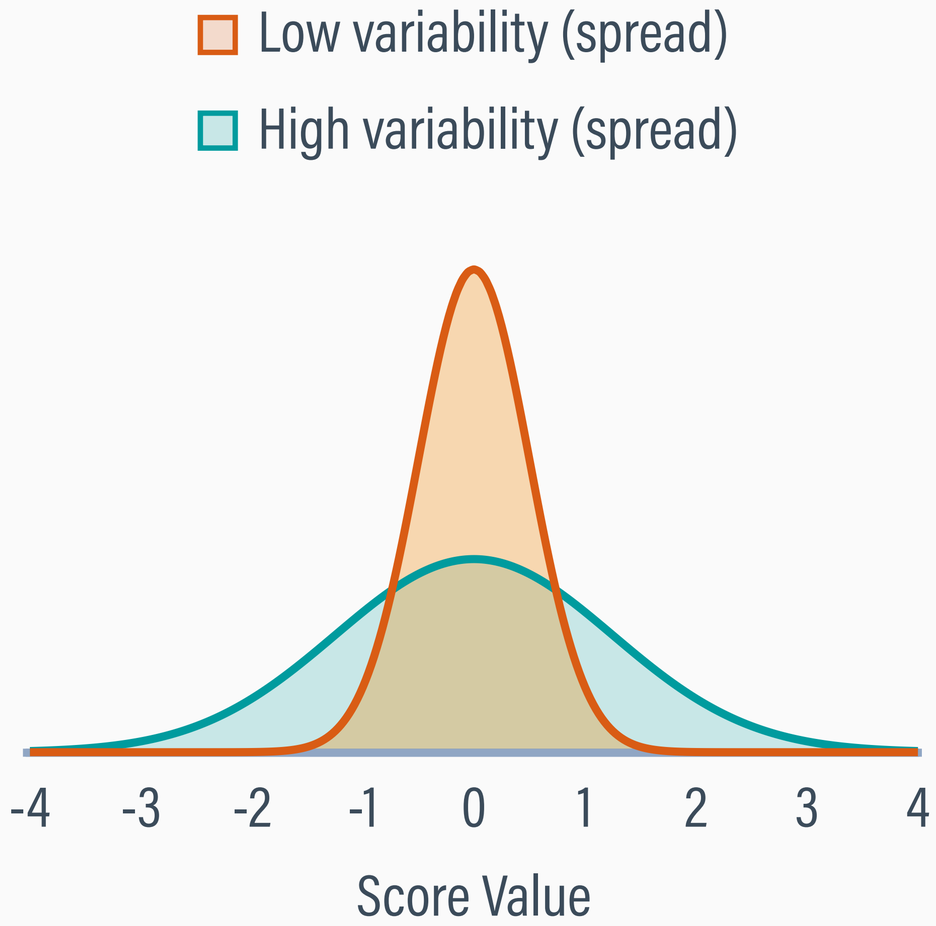
1. Measures of spread: Range, standard deviation, and variance
2. Statistical properties of the variance and standard deviation
3. Psychology application: Smoking cessation clinical trial
4. Standardizing: Expressing scores in standard deviation units
5. Study questions

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1. Measures of spread: Range, standard deviation, and variance
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VARIABILITY OR SPREAD

- Variability refers to the degree to which scores differ
- Low variability implies that scores are similar, whereas high variability means scores are different



MEASURES OF SPREAD

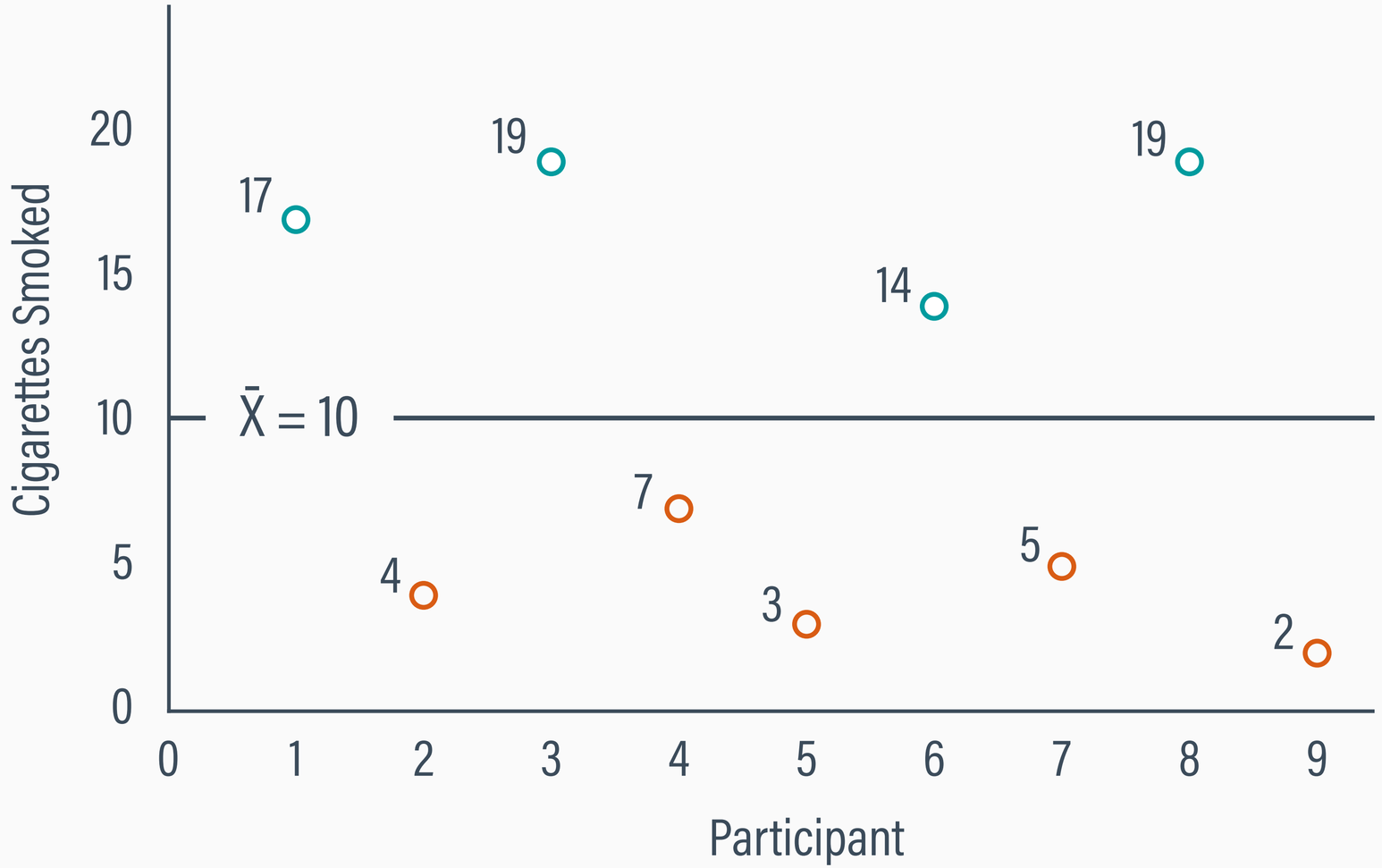
- Three definitions of variation: high vs. low score (the range), average distance from the scores to the mean (average absolute deviation and standard deviation), and average squared distance from the scores to the mean (the variance)
- The standard deviation is preferable for describing data, but the variance appears in formulas later in the class (t-tests, ANOVA)
- The variance is just the square of the standard deviation

SMALL SAMPLE ILLUSTRATION

- Later in the slides, we will apply measures of variability to a smoking cessation clinical trial
- To illustrate how variation is defined and computed, I use a small sample where participants report the typical number of cigarettes that they smoked in the last week

X
17
4
19
7
3
14
5
19
2

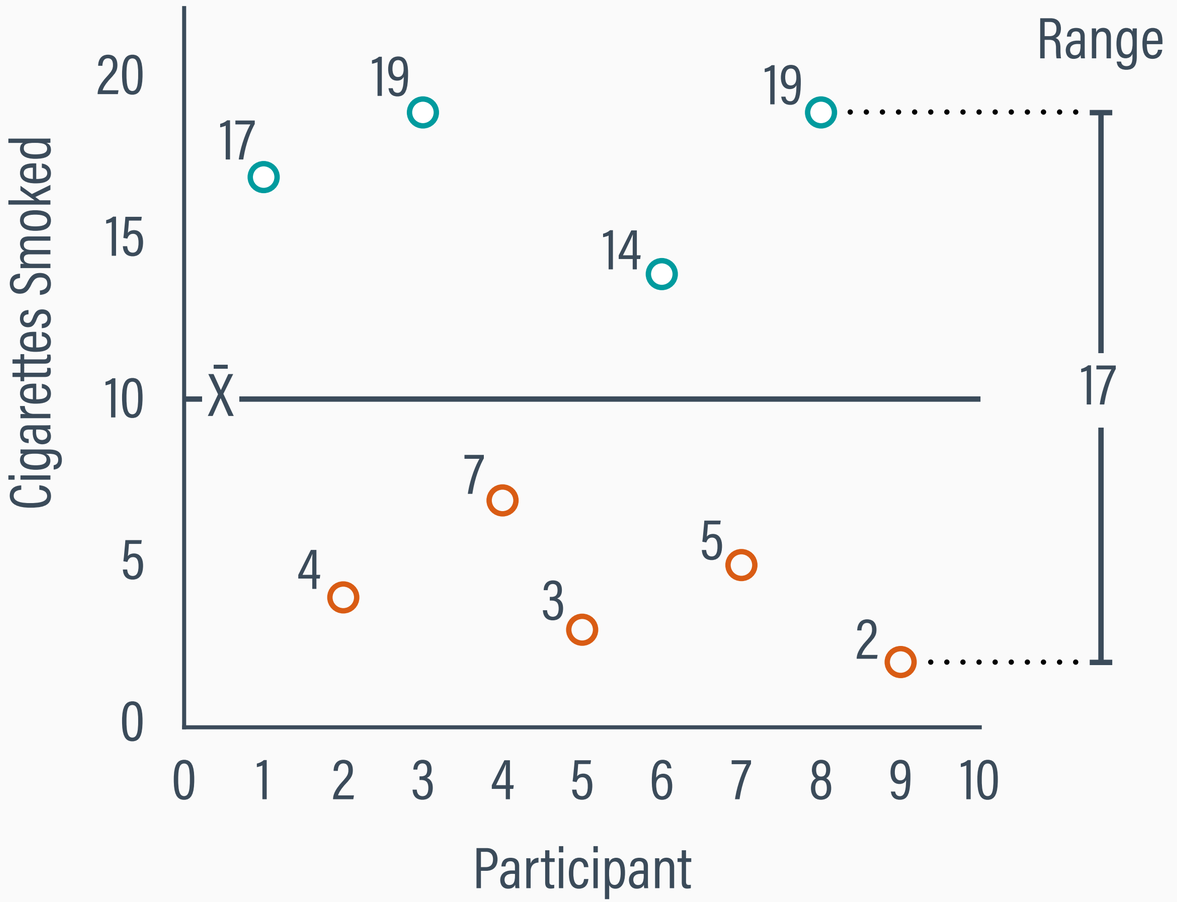
ILLUSTRATIVE DATA



X	$\bar{X}$
17	10
4	10
19	10
7	10
3	10
14	10
5	10
19	10
2	10

THE RANGE

- The range is the distance between the highest and lowest score
- Range = 19 – 2 = 17 cigarettes
- The range is not useful or reliable because it relies solely on two data points



STANDARD DEVIATION

- The standard deviation is average distance between the scores and the mean
- The average absolute deviation is a second measure that is uncommon in applications
- We will rely exclusively on the standard deviation because it is by far the predominant approach in practice

VARIATION RECIPE

1. Express scores as distances (deviations) from the mean
2. Square distances to eliminate negative values
3. Sum the squared distances
4. Average the squared distances to get the variance
5. Take the square root to get the standard deviation, which is the average distance from the mean

1. DISTANCE (DEVIATION) SCORES

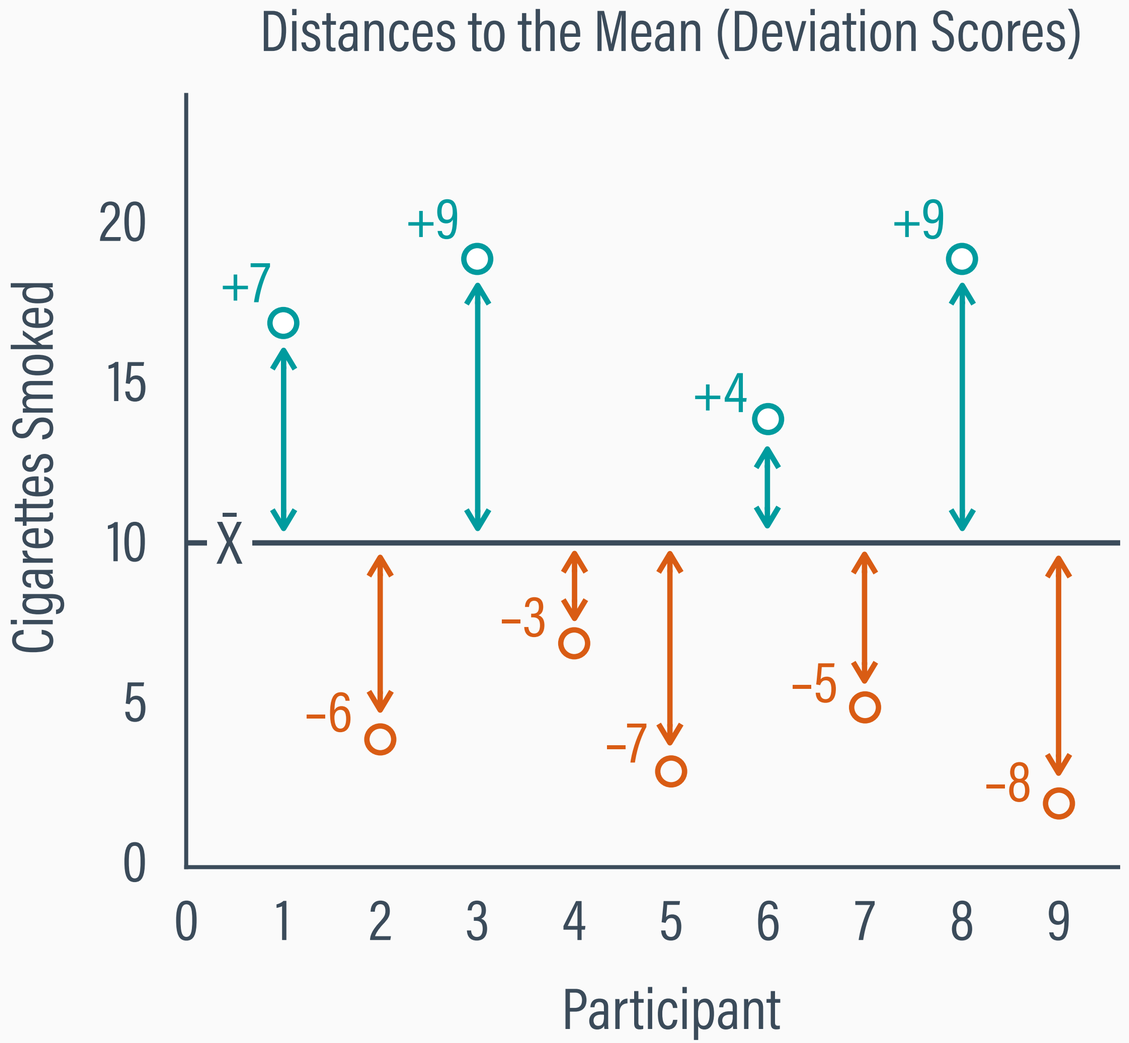
- The distance between a score X and the sample mean $\bar{X}$ is called a deviation score

$$d = X - \bar{X}$$

- Distance scores can be positive (score > mean), negative (score < mean), or zero (score = mean)

DEVIATION SCORES

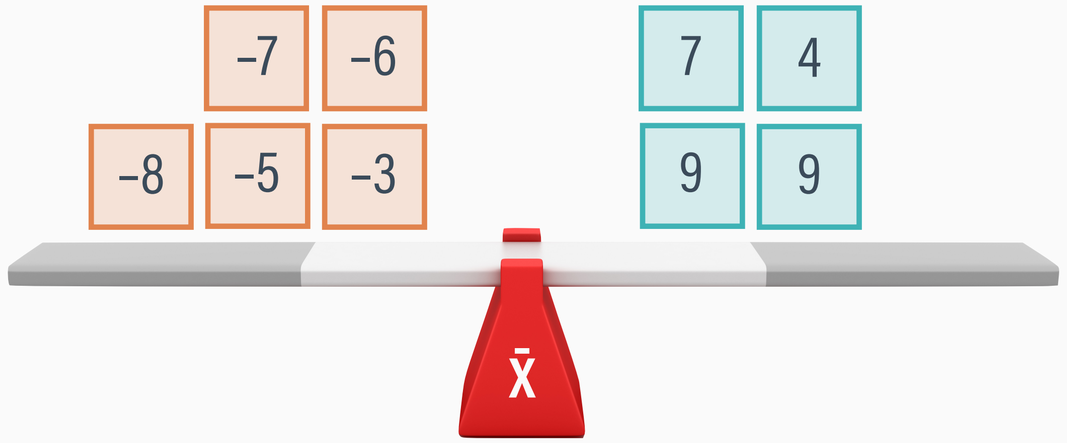
X	$\bar{X}$	d
19	10	9
19	10	9
17	10	7
14	10	4
7	10	-3
5	10	-5
4	10	-6
3	10	-7
2	10	-8



DEVIATION SCORES AVERAGE TO ZERO

X	$\bar{X}$	d
19	10	9
19	10	9
17	10	7
14	10	4
7	10	-3
5	10	-5
4	10	-6
3	10	-7
2	10	-8
	Sum =	**0**

- We cannot average deviation scores directly because the positive and negative distances from the mean always cancel out and must sum to zero



2. SQUARE DEVIATION SCORES

- Squaring distance scores eliminates negative values, so distances no longer cancel out

$$d^2 = (X - \bar{X})^2$$

- Achieves the same goal as the absolute value

X	$\bar{X}$	d	d^2
19	10	9	81
19	10	9	81
17	10	7	49
14	10	4	16
7	10	-3	9
5	10	-5	25
4	10	-6	36
3	10	-7	49
2	10	-8	64

3. SUM OF SQUARED DISTANCES

- Summing the squared distance scores gives a foundational measure called sum of squares (SS)

$$SS = \sum (X - \bar{X})^2 = \text{sum of squared distances}$$

- The sum of squares expresses the total amount of variability in the data as a lump sum

X	$\bar{X}$	d	d^2
19	10	9	81
19	10	9	81
17	10	7	49
14	10	4	16
7	10	-3	9
5	10	-5	25
4	10	-6	36
3	10	-7	49
2	10	-8	64
Sum of Squares =			**410**

4. AVERAGE SQUARED DISTANCE

- Averaging the squared distances between the scores and the mean gives a measure of variability called the variance

! This version of the formula is biased

$$s^2_{biased} = \frac{\sum(X - \bar{X})^2}{N} = \frac{SS}{N}$$

- The variance is just the arithmetic average (a sum divided by the number of observations) applied to squared scores

THE VARIANCE

- The average squared distance from the smoking scores to the mean is 45.6

$$s^2_{biased} = \frac{\sum(X - \bar{X})^2}{N} = \frac{SS}{N} = \frac{410}{9} = 45.6$$

- The variance will appear in equations later in the course, but we will not use it to describe data

X	$\bar{X}$	d	d^2
19	10	9	81
19	10	9	81
17	10	7	49
14	10	4	16
7	10	-3	9
5	10	-5	25
4	10	-6	36
3	10	-7	49
2	10	-8	64
		Sum of squares =	410

Variance (average of squared distances) = 45.6

5. AVERAGE DISTANCE TO THE MEAN

- Taking the square root undoes the squared distances, giving the average distance from the scores to the mean

! This version of the formula is biased

$$s_{biased} = \sqrt{\frac{\sum(X - \bar{X})^2}{N}} = \sqrt{s^2_{biased}}$$

- The standard deviation is average (typical) distance

THE STANDARD DEVIATION

- The standard deviation is the square root of the variance

$$s_{biased} = \sqrt{s^2_{biased}} = \sqrt{45.6} = 6.75$$

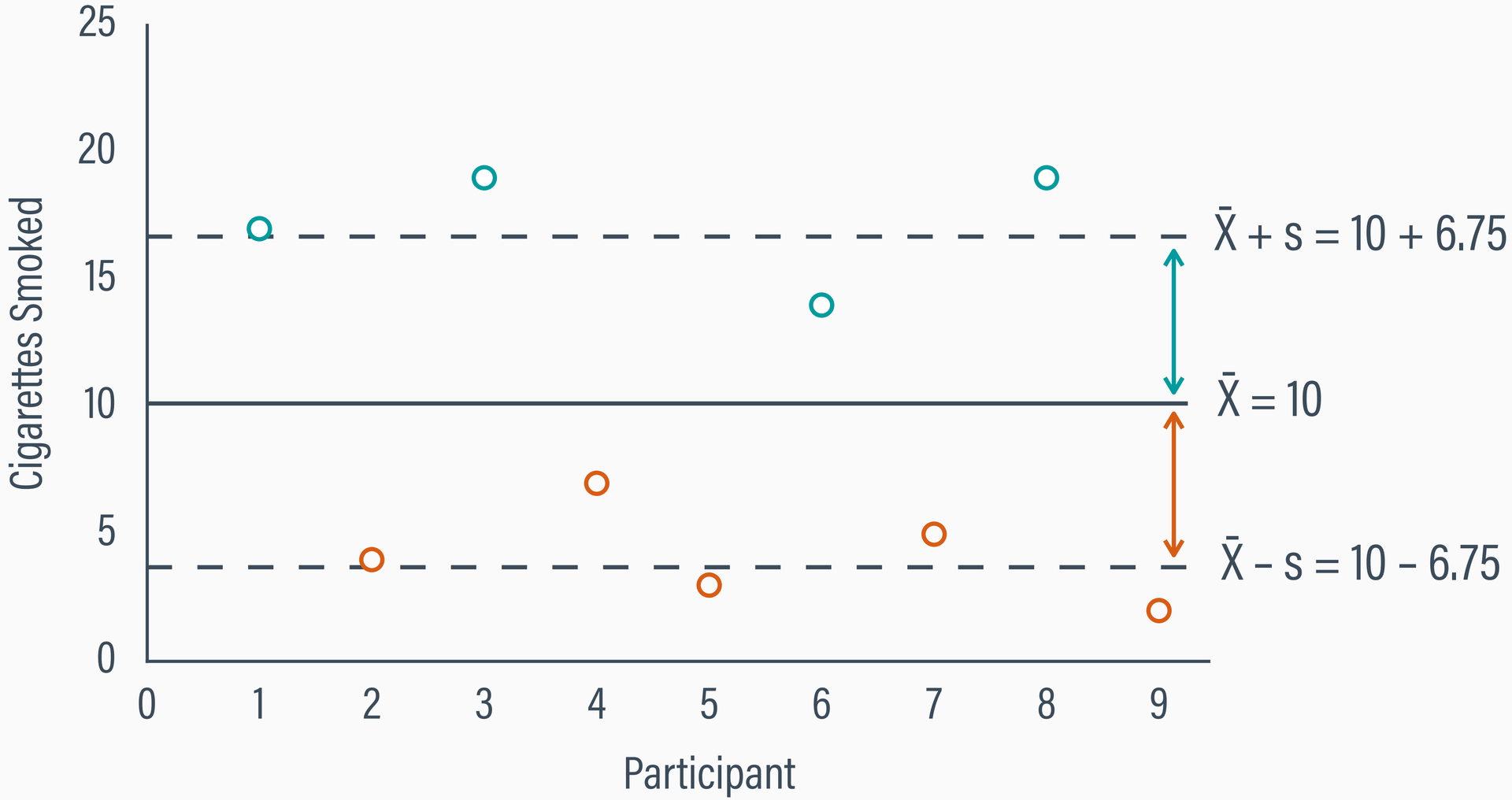
- The average (typical) distance from the smoking scores to the mean is 6.75 cigarettes

X	$\bar{X}$	d	d^2
19	10	9	81
19	10	9	81
17	10	7	49
14	10	4	16
7	10	-3	9
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4	10	-6	36
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2	10	-8	64

Sum of squares = 410

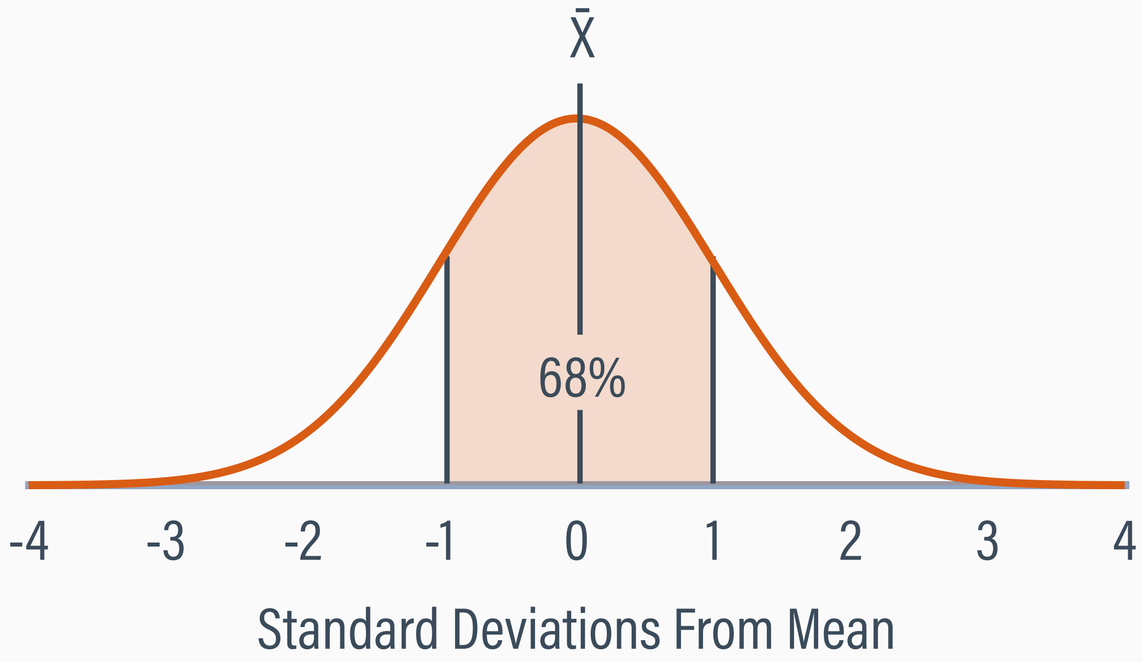
Variance (average of squared distances) = 45.6

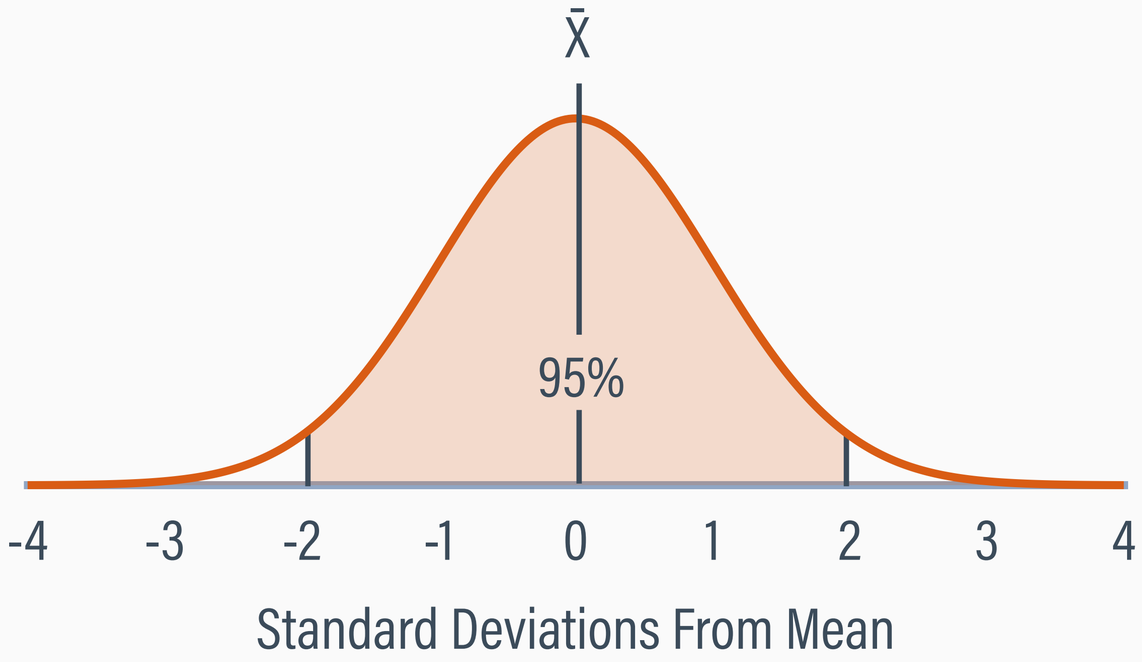
VISUALIZING THE STANDARD DEVIATION



RULE OF THUMB FOR NORMAL DATA

- In a normal curve, 68% of the scores are within ± 1 standard deviation of the mean, and 95% are within ≈ ± 2 standard deviations





OUTLINE

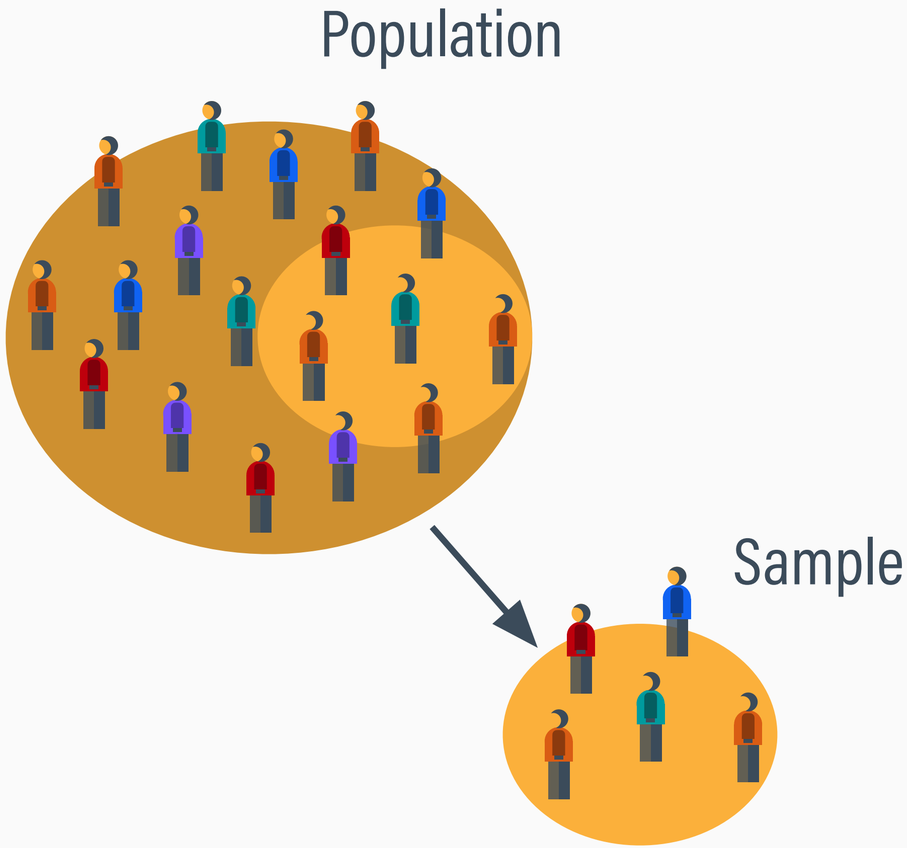
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IMPORTANT TERMINOLOGY

Participants	Statistic
Population = all possible participants who share an attribute of interest	Parameter = a hypothetical statistic computed using the full population
Sample = the subset of people who participated in the study	Estimate = an observed statistic computed from the sample data

POPULATIONS VS. SAMPLES

- A population is the entire group of individuals that are of interest in a study
- Researchers almost exclusively work with a smaller subset called a sample
- The usual goal is use a sample statistic as a best guess about the population statistic

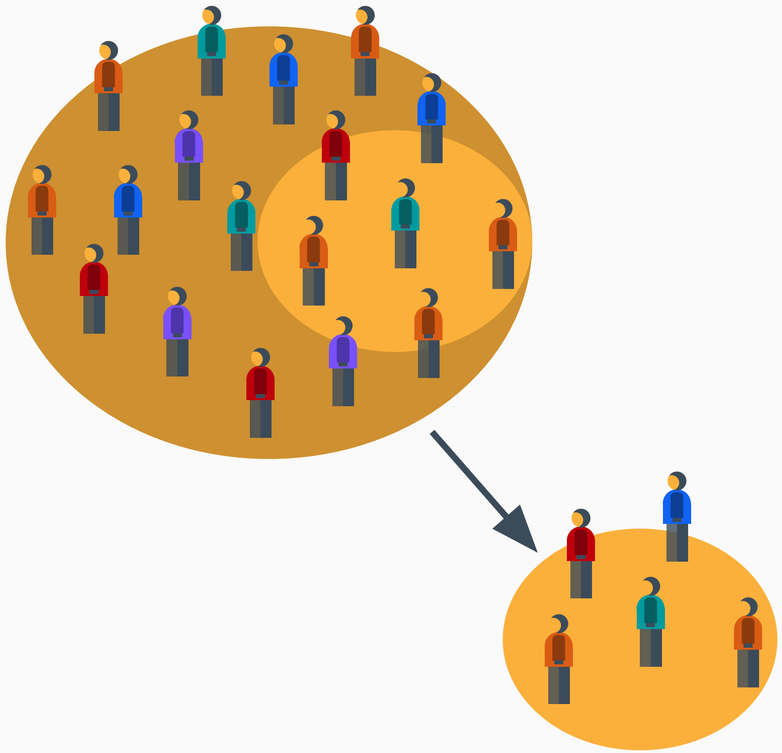


PARAMETERS VS. ESTIMATES

- Greek letters σ and σ^2 denote the true population-level standard deviation and variance (parameters)
- Roman letters s and s^2 reference a sample's standard deviation and variance (estimates)
- Ideally, the sample statistics will be close to the true values, but they will always differ

Population variance = σ^2

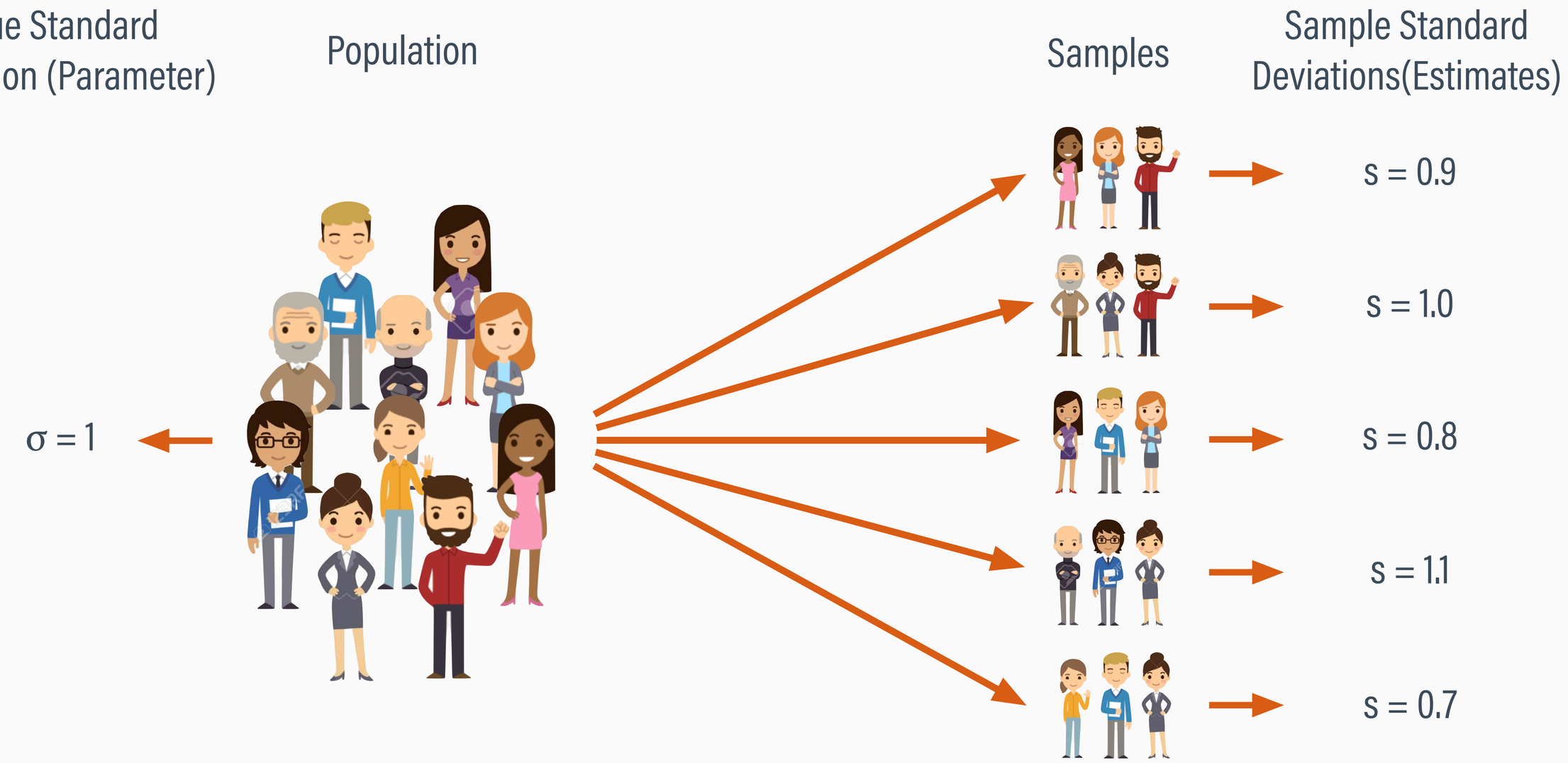
Population standard deviation = σ



Sample variance = s^2

Sample standard deviation = s

SAMPLE ESTIMATES DIFFER FROM TRUE VARIATION



BIASEDNESS OF SAMPLE ESTIMATES

- Like the mean, the standard deviation and variance will vary from sample to sample
- The earlier equations give biased estimates of the true population variation: on average (across many samples), estimates systematically miss in one direction
- On average, sample estimates will be too low (they underestimate the true variation in the population)

BIASED ESTIMATES

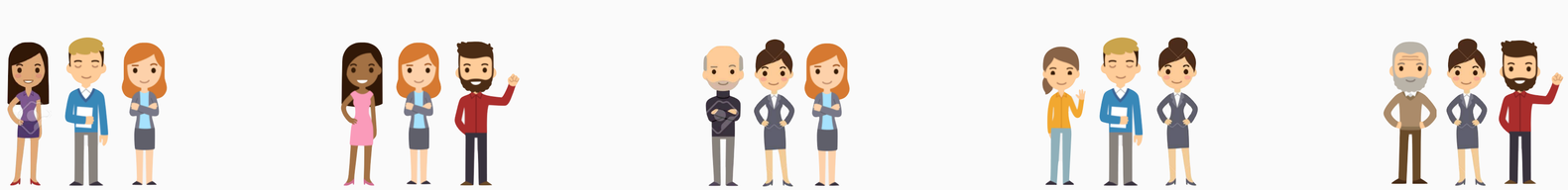
Estimates are systematically too low



Population

$\sigma = 1$

0.7 — 0.8 — 0.9 — 1 — 1.1 — 1.2 — 1.3



$s = 0.7$ $s = 0.8$ $s = 0.9$ $s = 1.0$ $s = 1.1$

Samples

SOURCE OF BIAS

Most of the time, the distance from the sample mean is smaller than the true mean.

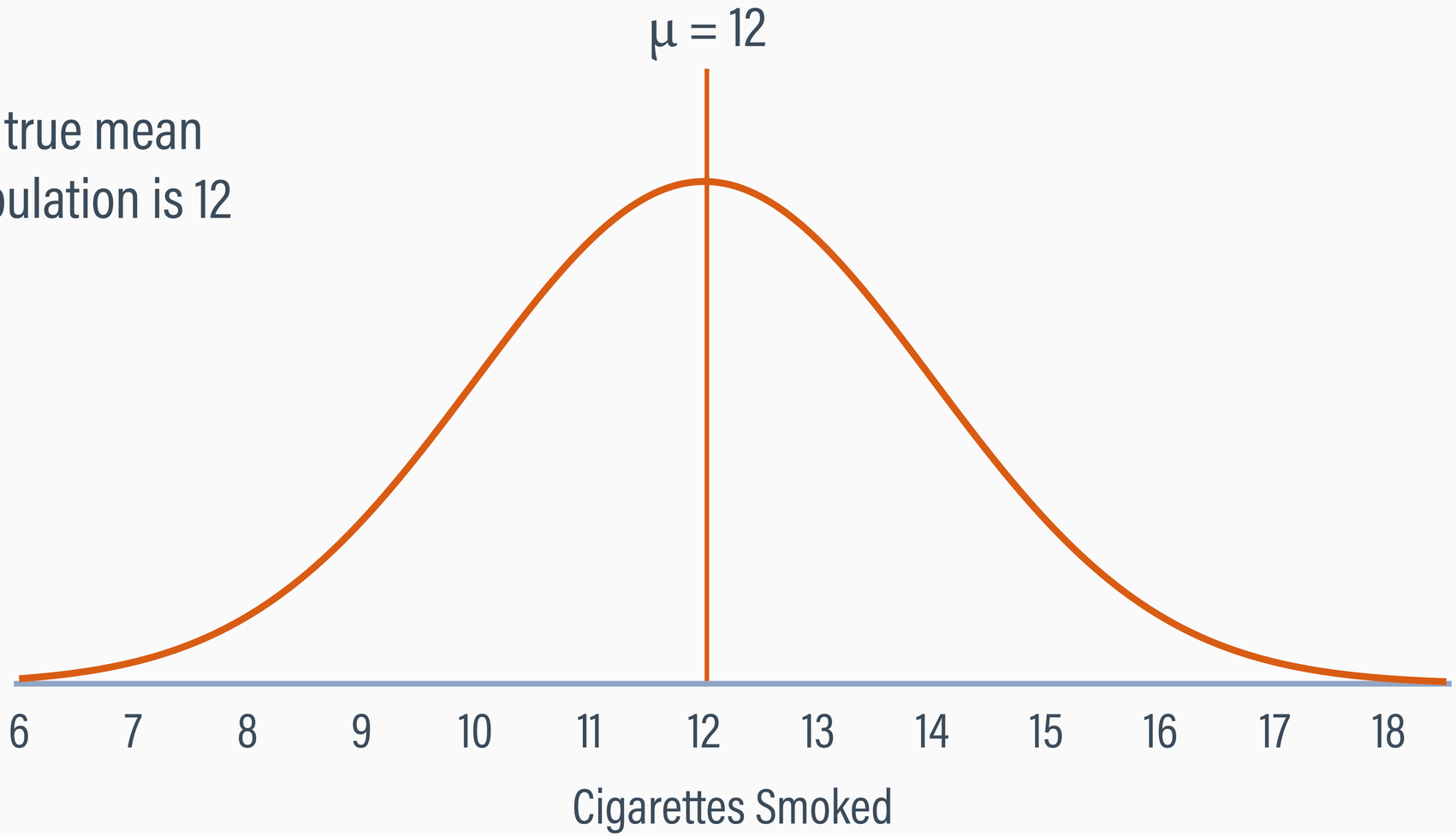
- We never know the true mean μ, so we must instead use the sample mean $\bar{X}$ to compute distance (deviation) scores

- Because the sample mean is the balancing point in the distribution, the sum of squared distances from $\bar{X}$ will be smaller than distances from any other value (including μ)

- This makes the numerator (sum of squares) too small, which reduces the value of the standard deviation and variance

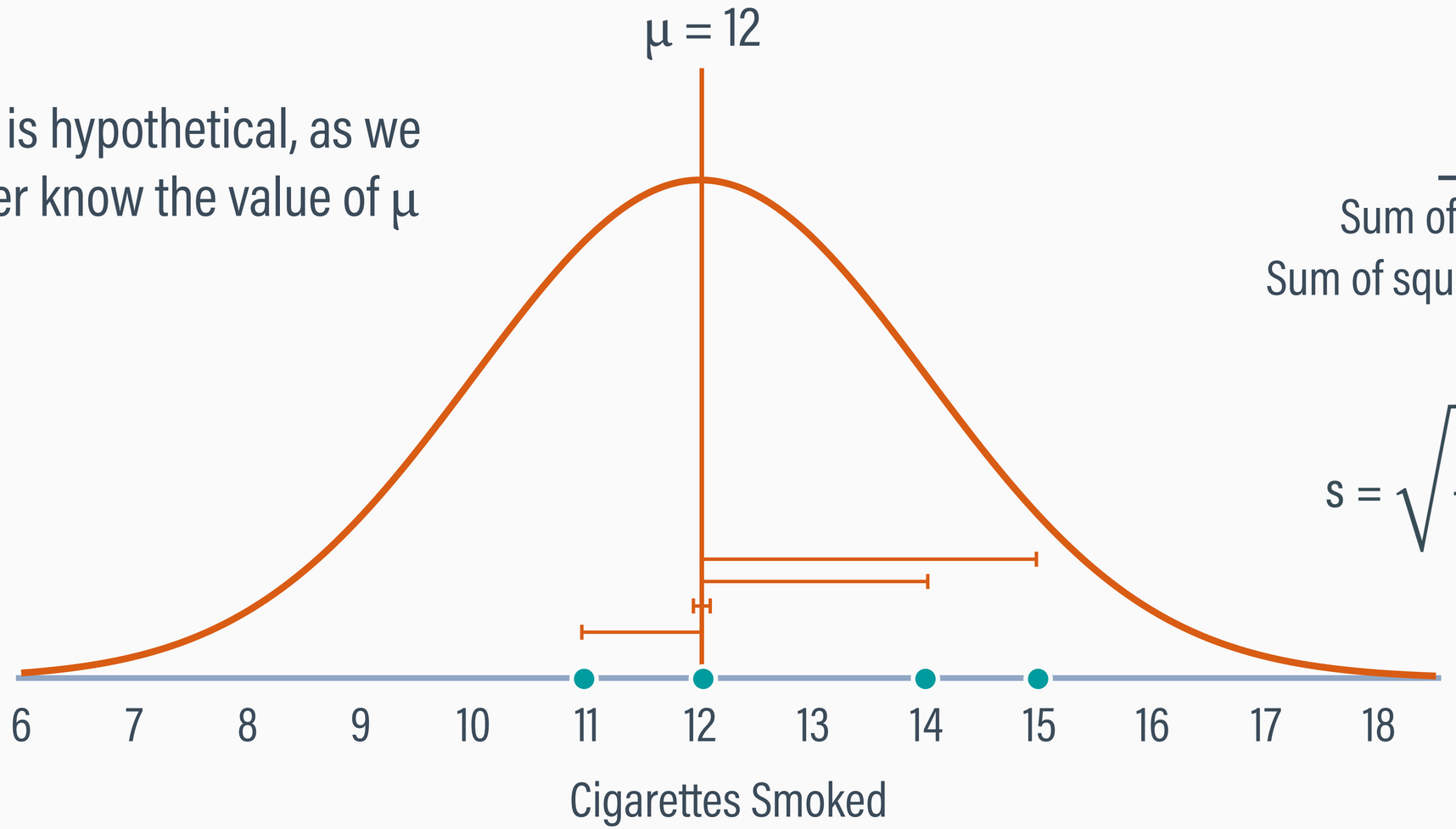
ILLUSTRATION

Pretend the true mean
in the full population is 12



VARIABILITY RELATIVE TO TRUE MEAN

This is hypothetical, as we never know the value of μ



X	μ	d	d^2
11	12	-1	1
12	12	0	0
14	12	2	4
15	12	3	9

Sum of distances = 4

Sum of squared distances = **14**

$$s = \sqrt{\frac{\sum(X - \mu)^2}{N}} = \sqrt{\frac{14}{4}} = 1.87$$

WHAT THE MATH SAYS

- Across persons, the expected value (average) of a squared deviation score is the variance

$$E\left[(X - \mu)^2\right] = \sigma^2$$

- Across many samples, the expected value (average) of the sum of squares is $N\sigma^2$

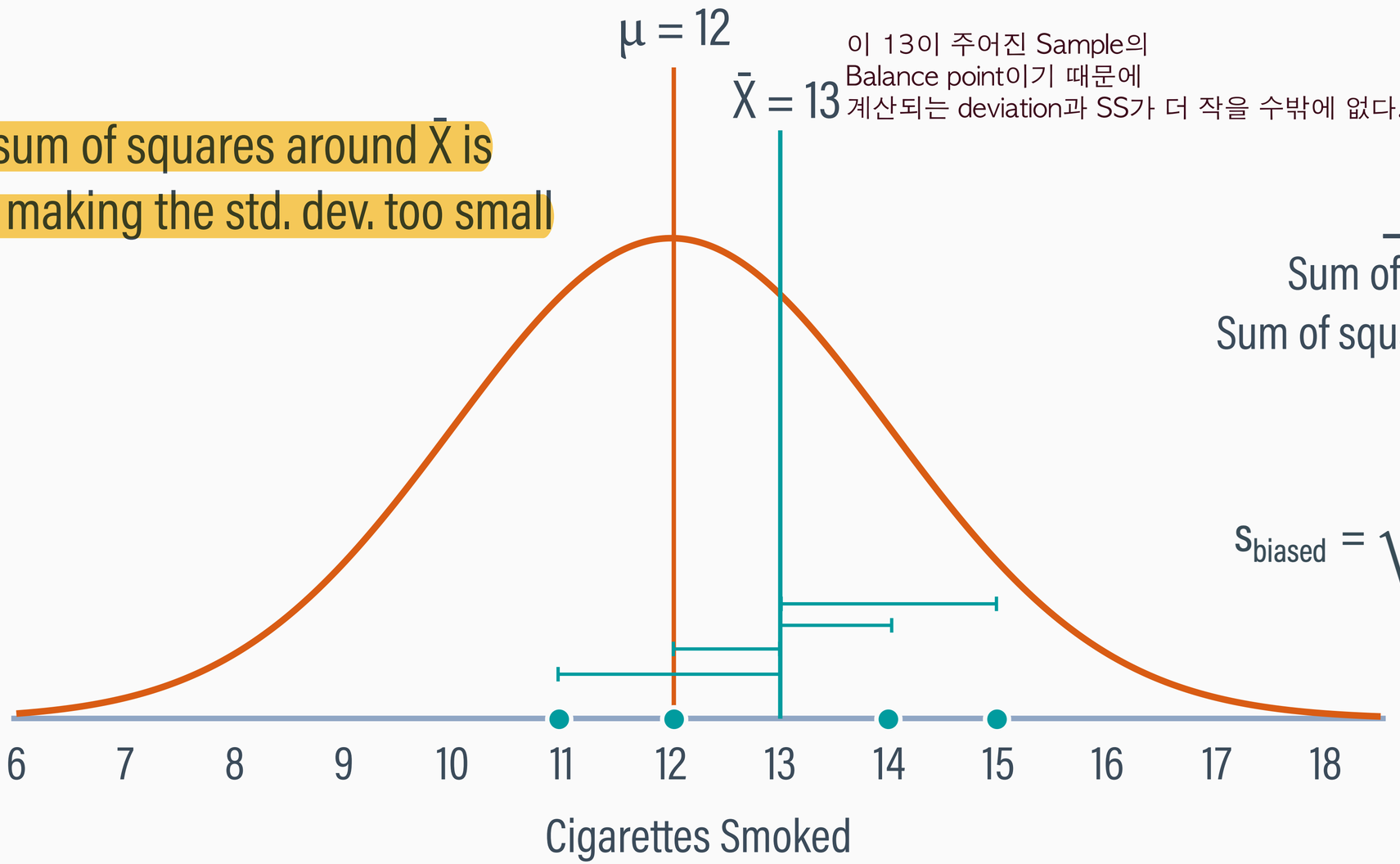
$$E[SS] = E\left[\sum (X - \mu)^2\right] = N\sigma^2$$

- If we knew the true mean, dividing by N would give an unbiased estimate of the variance

$$\frac{E[SS]}{N} = \frac{N\sigma^2}{N} = \sigma^2$$

VARIABILITY RELATIVE TO SAMPLE MEAN

The sum of squares around $\bar{X}$ is smaller, making the std. dev. too small



이 13이 주어진 Sample의
Balance point이기 때문에
계산되는 deviation과 SS가 더 작을 수밖에 없다.

X	$\bar{X}$	d	d^2
11	13	-2	4
12	13	-1	1
14	13	1	1
15	13	2	4

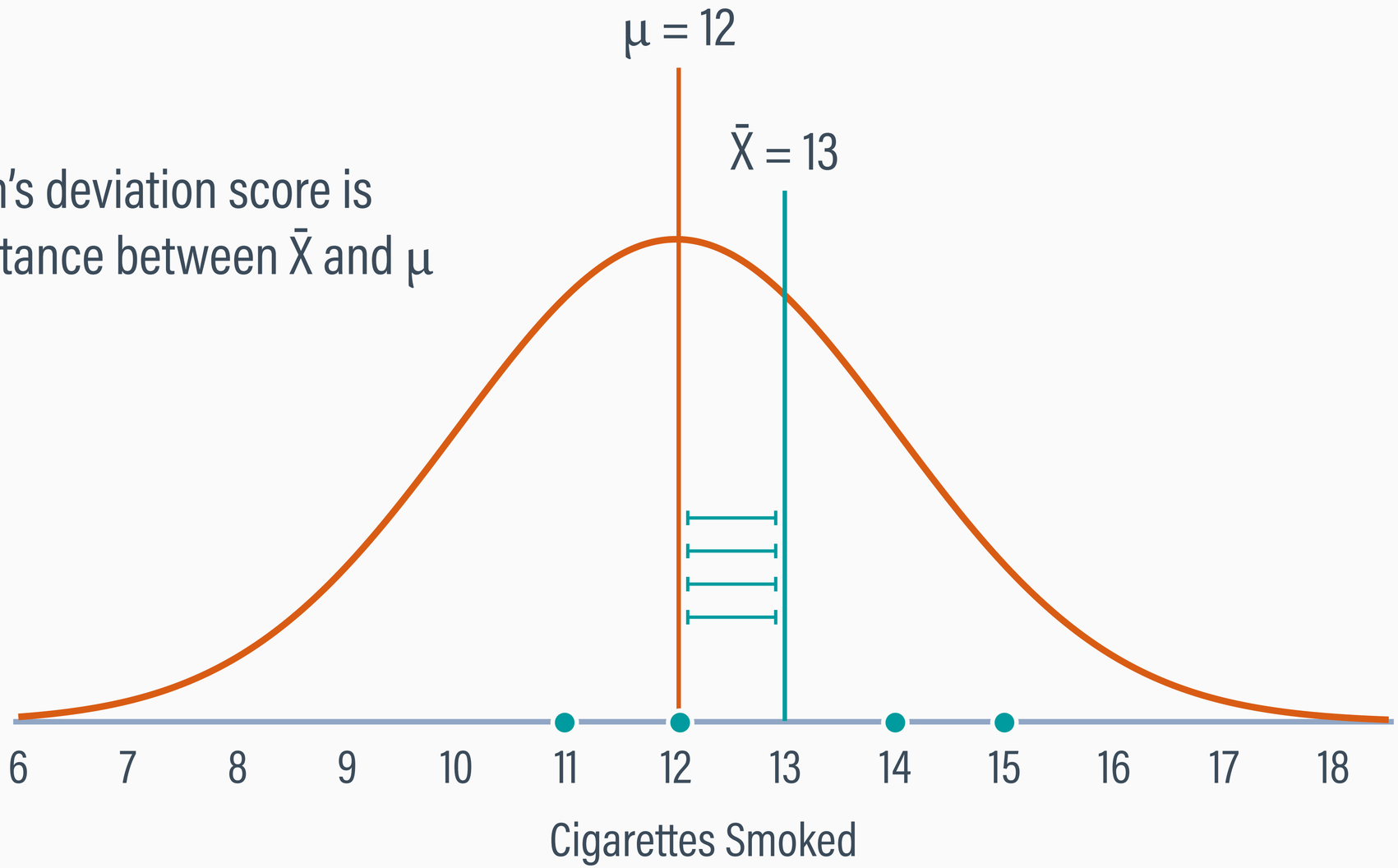
Sum of distances = 0

Sum of squared distances = **10**

$$s_{biased} = \sqrt{\frac{\sum(X - \bar{X})^2}{N}} = \sqrt{\frac{10}{4}} = 1.58$$

MISSING DISTANCES

Each person's deviation score is missing the distance between $\bar{X}$ and μ



WHAT THE MATH SAYS

Missing some distances
오 대박이당
그래서 무조건 작을 수밖에 없구나

- The sum of squares around μ consists of two distances, one of which we can't have: $(\bar{X} - \mu)^2$
- Thus, the sum of squares computed using $\bar{X}$ is less than the sum of squares based on μ
- If the sum of squares is too small, dividing by N will systematically underestimate variation

$$\sum(X - \mu)^2 = \sum(X - \bar{X})^2 + N(\bar{X} - \mu)^2$$

$$E\left[\sum(X - \bar{X})^2\right] < E\left[\sum(X - \mu)^2\right]$$

UNBIASED ESTIMATES

- Averaging the squared distances by N – 1 (the degrees of freedom) corrects the downward bias in the sum of squares

$$s^2 = \frac{\sum(X - \bar{X})^2}{N - 1} = \frac{SS}{N - 1} \qquad s = \sqrt{\frac{\sum(X - \bar{X})^2}{N - 1}} = \sqrt{s^2}$$

- On average (across many samples), averaging distances by N – 1 recovers the true population variation

WHAT THE MATH SAYS, CONTINUED

- Derivation gives the expected value (average) of the sum of squares around $\bar{X}$
- What value would we need to divide by for the sum of squares around the sample mean to recover the population variance, on average?
- On average, dividing by N – 1 recovers σ^2

This equals to variance of sample mean, which equals to missing distances

$$E\left[\sum(X-\bar{X})^2\right] = E\left[\sum(X-\mu)^2\right] - E\left[N(\bar{X}-\mu)^2\right]$$

$$E\left[\sum(X-\bar{X})^2\right] = N\sigma^2 - N\frac{\sigma^2}{N} = (N-1)\sigma^2$$

$$\frac{E\left[\sum(X-\bar{X})^2\right]}{?} = \frac{(N-1)\sigma^2}{?} = \sigma^2$$

$$? = (N-1)$$

DEGREES OF FREEDOM

- The N – 1 term is called the degrees of freedom
- The degrees of freedom is the adjusted sample size needed to get an unbiased estimate of variation correction term?
- Degrees of freedom adjustment terms are common in statistical formulas (e.g., t-tests, ANOVA, regression)

DF AS NUMBER OF UNIQUE DATA POINTS

- The degrees of freedom can also be understood as the number of unique data points that contribute to an estimate
- After estimating the sample mean (and plugging it into the sum of squares formula), we have N – 1 unique data points with which to estimate variation
- Using the sample mean in an equation reduces the amount of available information in the data

DEGREES OF FREEDOM ILLUSTRATION

- Suppose that a sample of N = 3 scores has a mean of $\bar{X} = 10$ (there are N – 1 = 2 degrees of freedom)
- Two of the scores (N – 1) can take on any value, but the third is fully determined after we estimate $\bar{X}$ (the amount of unique information available in the data has been reduced)
- If two of the scores are 8 and 11, then the third must be 11; if two of the scores are 12 and 17, then the third must be 1; if two of the scores are 8 and 9, then the third must be 13

THE VARIANCE

- The average squared distance from the smoking scores to the mean is 51.25

$$s^2 = \frac{\sum(X - \bar{X})^2}{N - 1} = \frac{SS}{N - 1} = \frac{410}{8} = 51.25$$

- The biased formula gave a smaller estimate (45.6) because it is systematically to low

X	$\bar{X}$	d	d^2
19	10	9	81
19	10	9	81
17	10	7	49
14	10	4	16
7	10	-3	9
5	10	-5	25
4	10	-6	36
3	10	-7	49
2	10	-8	64
		Sum of squares =	410

Variance (average of squared distances) = 51.25

THE STANDARD DEVIATION

- The average (typical) distance from the smoking scores to the mean is 7.16 cigarettes

$$s = \sqrt{s^2} = \sqrt{51.25} = 7.16$$

- The biased formula gave a smaller estimate (6.75) because it is systematically to low

쥐 9마리 데꼬 하는 실험이면 의미가 있겠지만… 한편으로는 의미가 없음
SEM에서는 절대 N-1로 나눌 일이 없다?!

X	$\bar{X}$	d	d^2
19	10	9	81
19	10	9	81
17	10	7	49
14	10	4	16
7	10	-3	9
5	10	-5	25
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2	10	-8	64

Sum of squares = 410

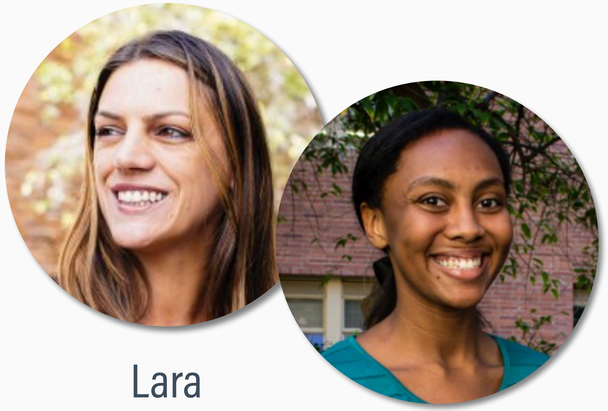
Variance (average of squared distances) = 51.25

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SMOKING AND DRINKING CESSATION TRIAL

Pharmacological treatments that can concomitantly address cigarette smoking and heavy drinking stand to improve health care delivery for these highly prevalent co-occurring conditions. This superiority trial compared the combination of varenicline and naltrexone against varenicline alone for smoking cessation and drinking reduction among heavy-drinking smokers.



Lara Ray

ReJoyce Green

Ray, L.A., Green, R., Enders, C., et al. (2021). Efficacy of combining varenicline and naltrexone for smoking cessation and drinking reduction: A randomized clinical trial. *American Journal of Psychiatry*, *178*, 818–828.

KEY VARIABLES



Breath (alveolar) carbon monoxide

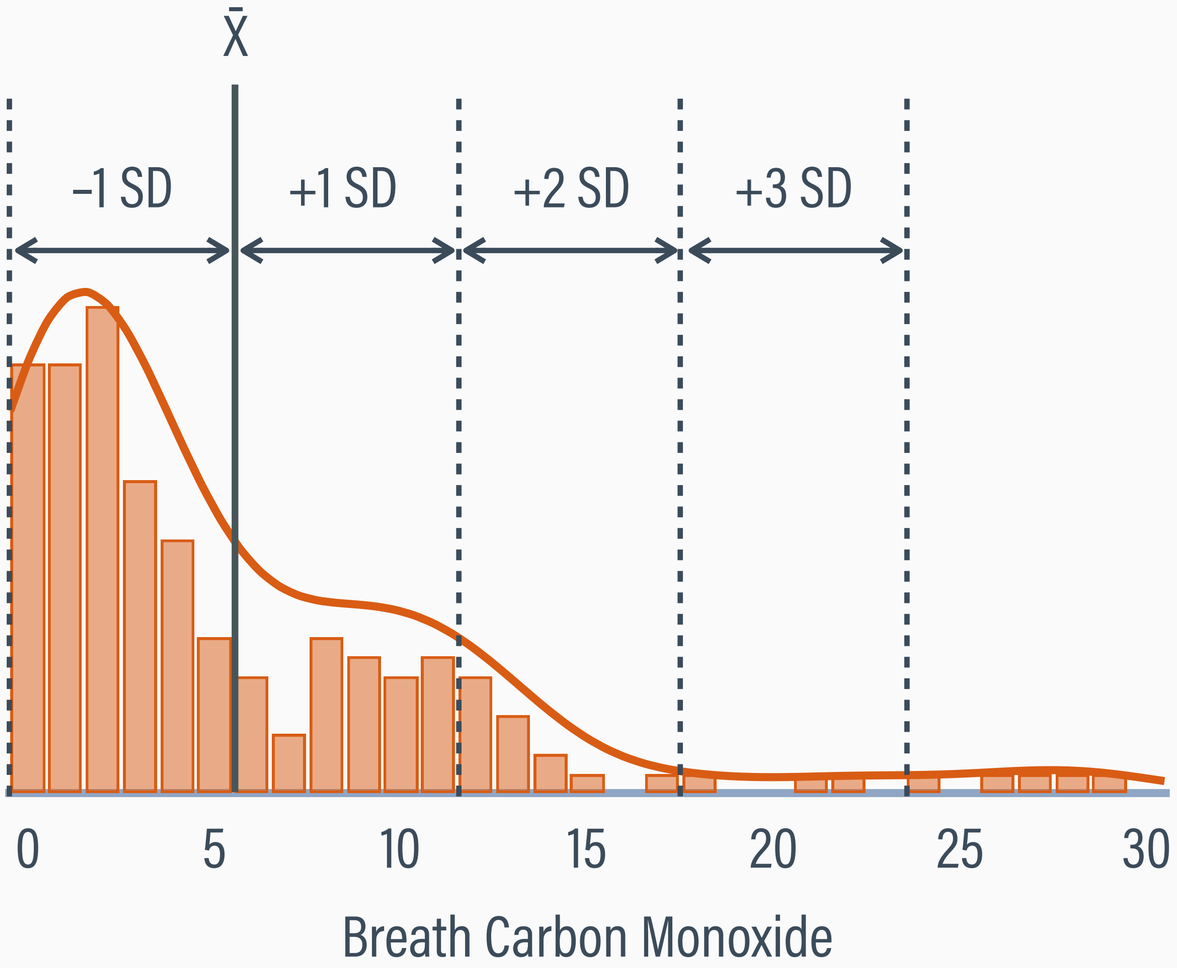
A measure of carbon monoxide in the lungs. Breath carbon monoxide is a biomarker of smoking behavior common in clinical trials. Higher scores reflect more frequent smoking.



Medication arm

Participants were randomly assigned to receive one of two meds: varenicline plus naltrexone or varenicline plus placebo pills

DESCRIPTIVE STATISTICS



Mean (X̄)	5.53
Std. Dev. (s)	5.96
N	165

R OUTPUT

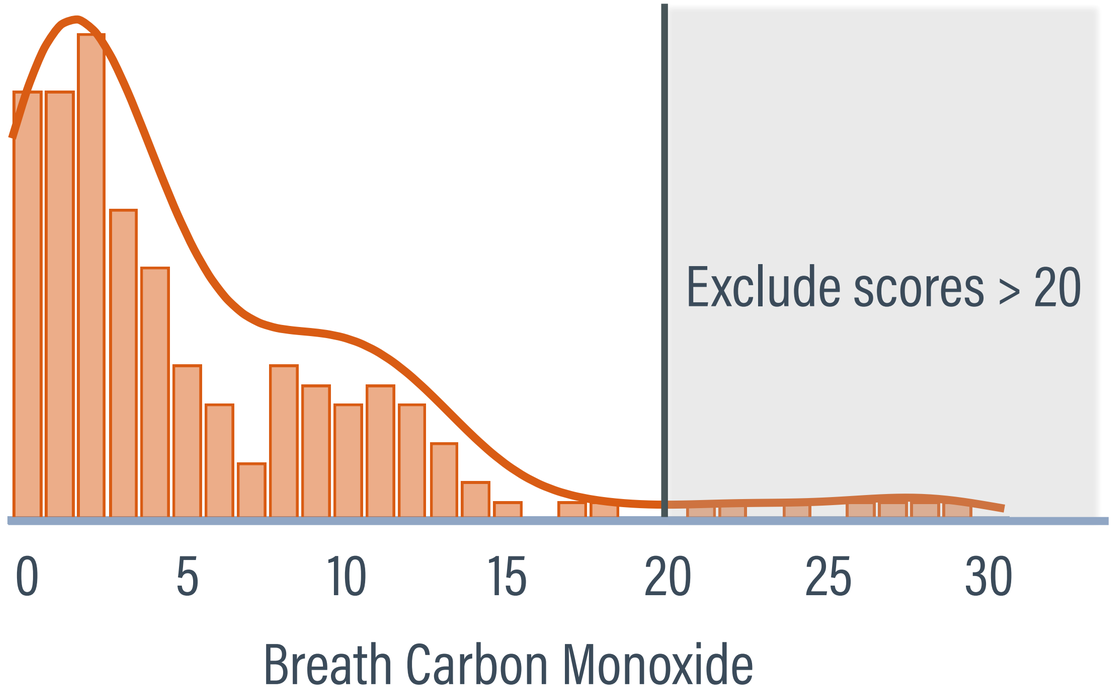
	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Participant	1	165	83.00	47.78	83	83.00	60.79	1	165	164	0.00	-1.22	3.72
Condition*	2	165	1.50	0.50	2	1.50	0.00	1	2	1	-0.01	-2.01	0.04
Gender*	3	165	1.61	0.49	2	1.63	0.00	1	2	1	-0.43	-1.83	0.04
COWeek0	4	165	10.55	6.86	9	9.82	5.93	0	46	46	1.41	3.60	0.53
COWeek4	5	165	5.46	5.10	4	4.69	4.45	0	24	24	1.24	0.96	0.40
COWeek8	6	165	5.53	5.96	3	4.55	2.97	0	29	29	1.75	3.35	0.46
QuitCigsWeek4*	7	165	1.62	0.49	2	1.65	0.00	1	2	1	-0.48	-1.78	0.04
QuitCigsWeek8*	8	165	1.64	0.48	2	1.68	0.00	1	2	1	-0.59	-1.66	0.04
DrinksWeek0	9	165	6.40	4.42	6	5.77	2.97	1	35	34	2.44	10.46	0.34
DrinksWeek4	10	165	3.59	2.98	3	3.30	2.97	0	13	13	0.80	0.30	0.23
DrinksWeek8	11	165	3.23	2.68	3	2.97	2.97	0	13	13	0.89	0.67	0.21
CigsWeek0	12	165	14.22	8.22	12	13.05	5.93	3	51	48	1.59	3.20	0.64
CigsWeek4	13	165	4.18	5.70	2	3.08	2.97	0	41	41	3.01	13.13	0.44
CigsWeek8	14	165	3.16	4.75	2	2.25	2.97	0	35	35	3.44	16.81	0.37



In small groups of two or three, interpret the standard deviation in practical terms. Your explanation should use the numeric value from the example, and it should be understandable to a non-technical audience without losing accuracy.

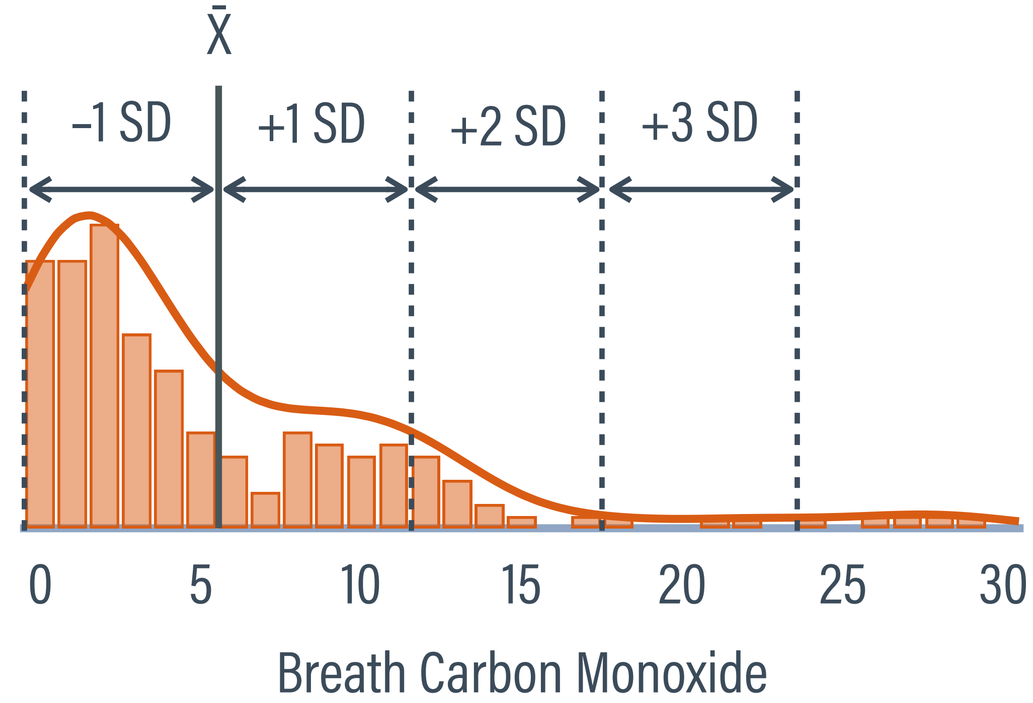


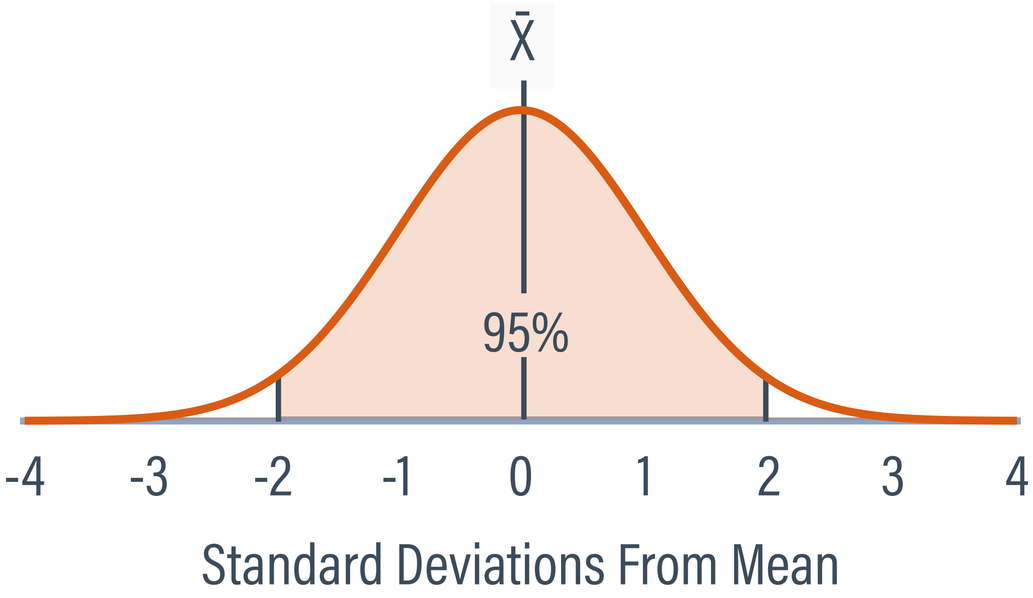
In small groups of two or three, discuss what would happen to the standard deviation if you excluded outlier participants with breath CO scores > 20?





In small groups of two or three, discuss the utility of the rule of thumb that 95% of scores fall within ± 2 standard deviations of the mean

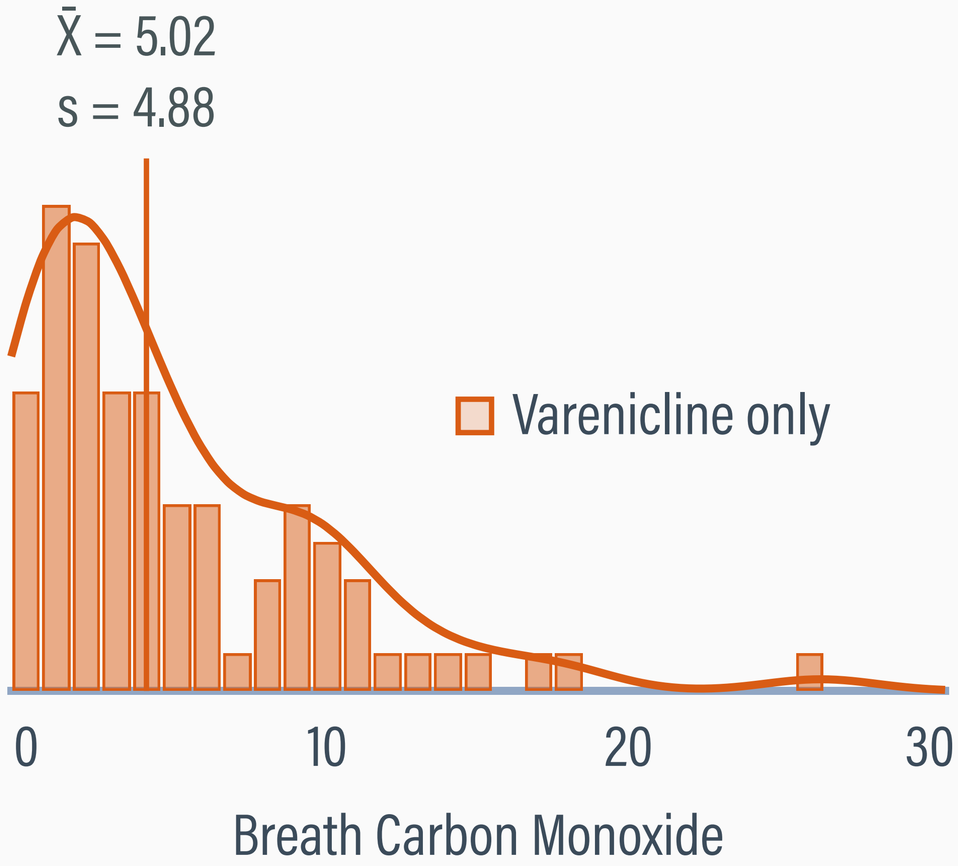


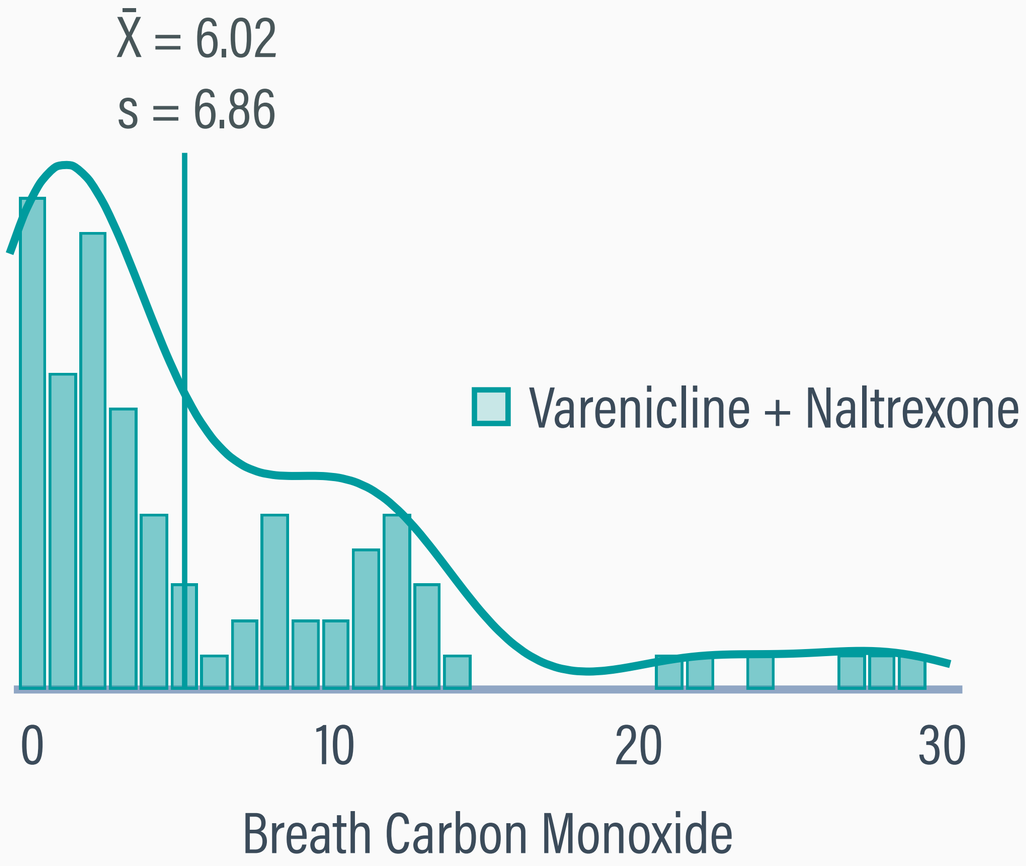


COMPARATIVE RESEARCH QUESTIONS

- Comparative research questions ask whether two or more groups (or occasions) differ from one another
- Question: Do participants in the two treatment groups differ in their smoking levels?
- We can answer this question by comparing descriptive statistics

GROUP STATISTICS



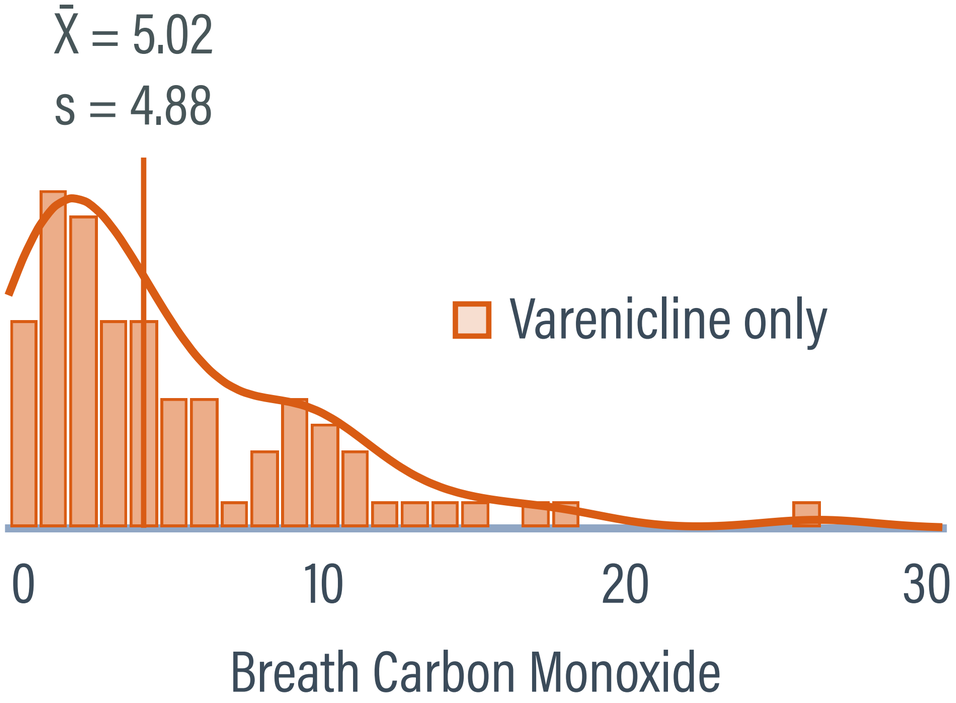


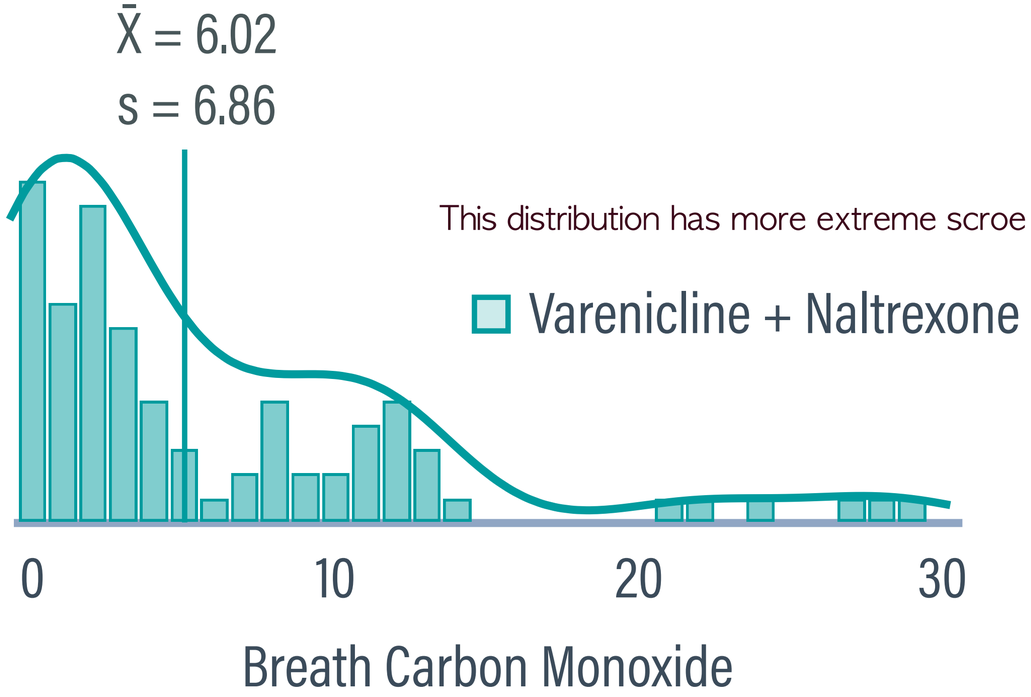
R OUTPUT

```
 Descriptive statistics by group
Condition: Varenicline
         vars  n mean   sd median trimmed  mad min max range skew kurtosis   se
COWeek8     1 82 5.02 4.88    3.5    4.33 3.71   0  26    26 1.57     3.09 0.54
------------------------------------------------------------------------------
Condition: Varenicline + Naltrexone
         vars  n mean   sd median trimmed  mad min max range skew kurtosis   se
COWeek8     1 83 6.02 6.86      3    4.81 4.45   0  29    29 1.62     2.31 0.75
```



In small groups of two or three, discuss the standard deviation differences. What feature of the distribution causes one group's standard deviation to be smaller than the other's?





OUTLINE

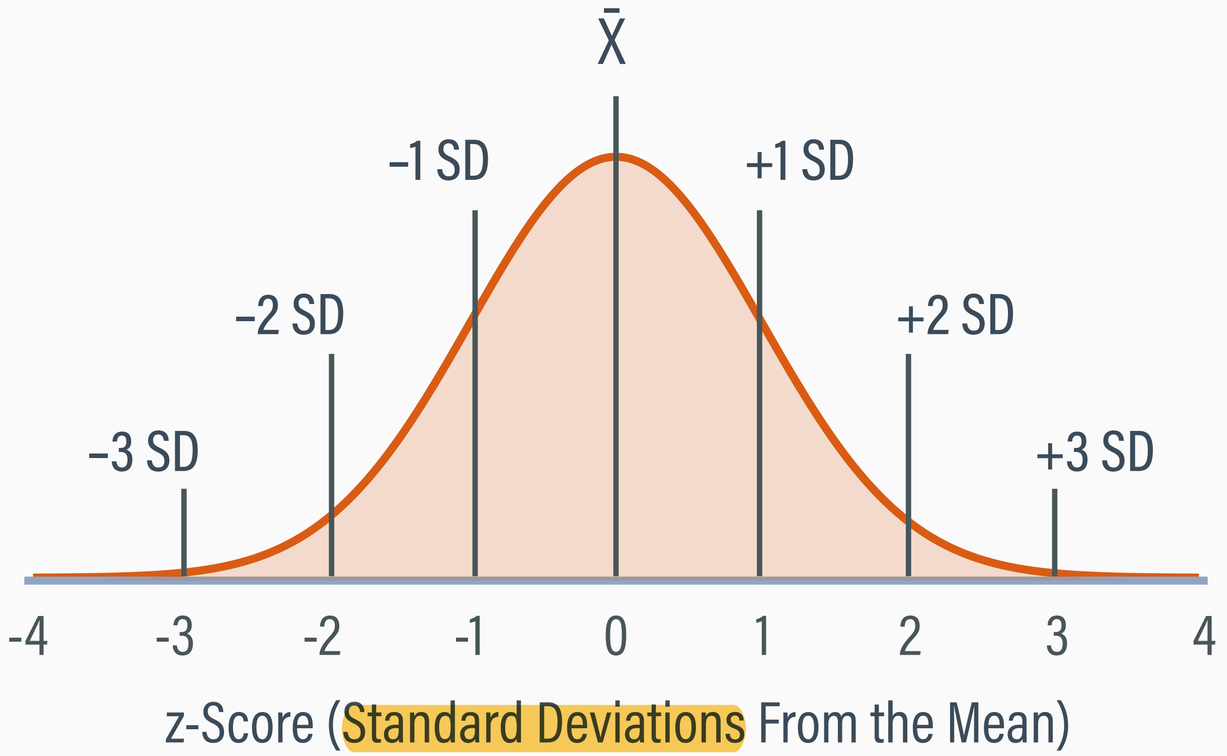
1. Measures of spread: Range, standard deviation, and variance
2. Statistical properties of the variance and standard deviation
3. Psychology application: Smoking cessation clinical trial
4. Standardizing: Expressing scores in standard deviation units
5. Study questions

STANDARDIZED SCORES

- Many physical variables (e.g., age, height, weight) have natural metrics that are easy to understand
- Psychological variables (e.g., depression, narcissism) are often use sum scores that do not have inherent metrics
- To facilitate interpretation, researchers often use standard scores with “universal” metrics (e.g., T-scores, IQ, GRE)

z-SCORES

- The z-score scale is a common standardized metric that expresses scores as standard deviation units from the mean

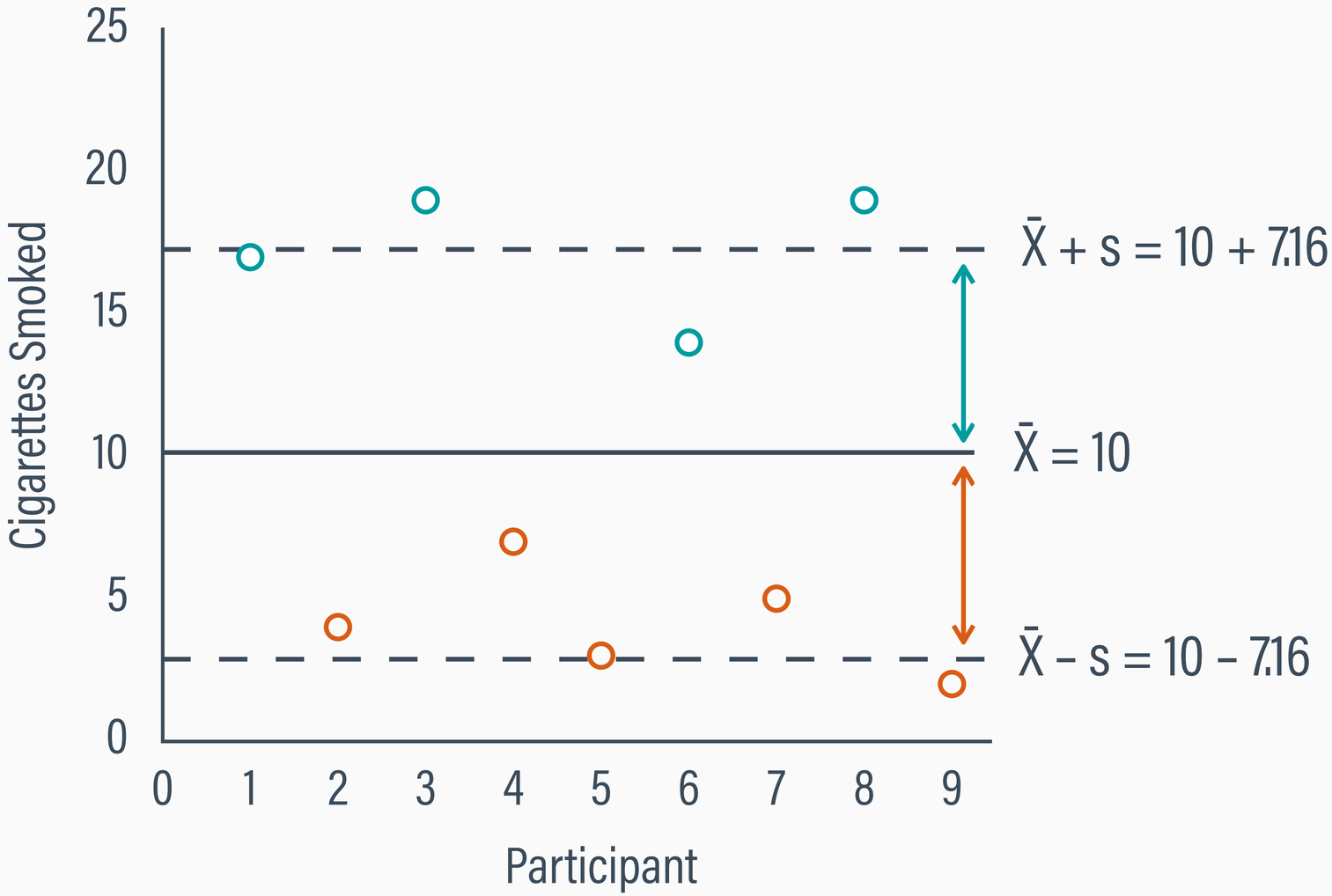


SMALL SAMPLE ILLUSTRATION

- To illustrate standardization, return to the earlier example where N = 9 participants reported the typical number of cigarettes that they smoked in the last week
- The mean was $\bar{X} = 10$, and the standard deviation was s = 7.16

X
19
19
17
14
7
5
4
3
2

VISUALIZING z-SCORES



Standard deviation units from the mean

X	z
19	1.26
19	1.26
17	0.98
14	0.56
7	−0.42
5	−0.7
4	−0.84
3	−0.98
2	−1.12

z-SCORE FORMULA

- The numerator captures the raw score's distance (deviation) from the sample mean $\bar{X}$

  This is the story 'inside' the sample. 너가 떠올렸던 것은 sample mean distribution이다.

$$z = \frac{X - \bar{X}}{s} = \frac{\text{distance from mean}}{\text{standard deviation}}$$

- Dividing by the standard deviation converts the distance to standard deviation units

FORMULA ILLUSTRATION

- A score of 17 is about 0.98 standard deviations above the sample mean ($\bar{X} = 10$), and a score of 4 is roughly 0.84 standard deviation units below the mean

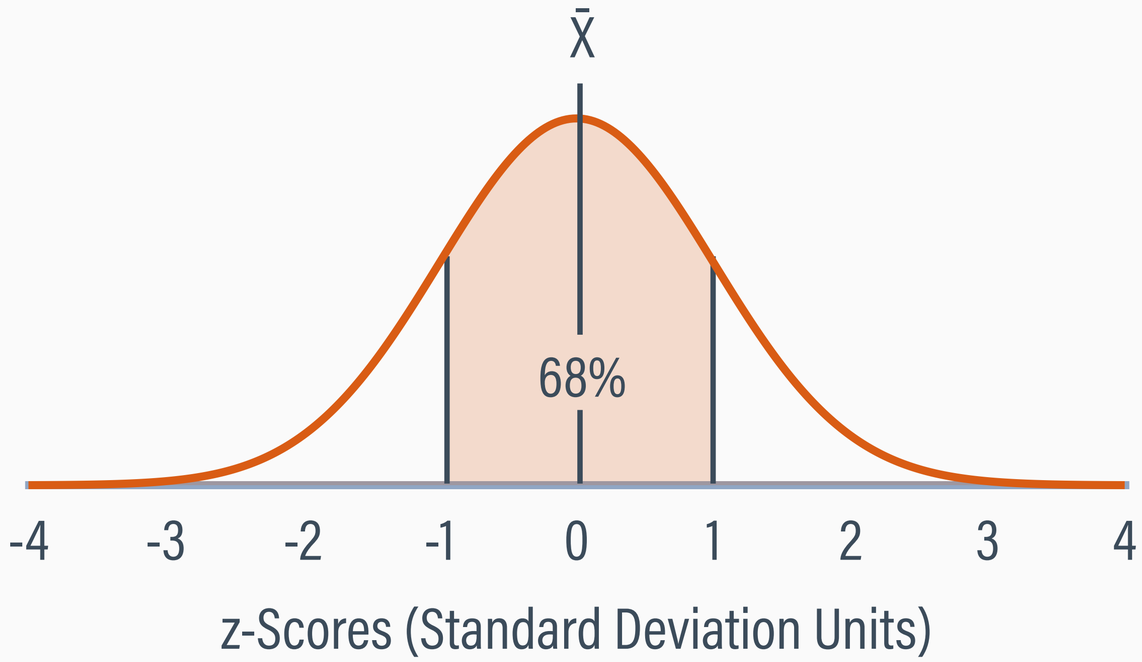
$$z = \frac{X - \bar{X}}{s} = \frac{17 - 10}{7.16} = \frac{+7}{7.16} = +0.98$$

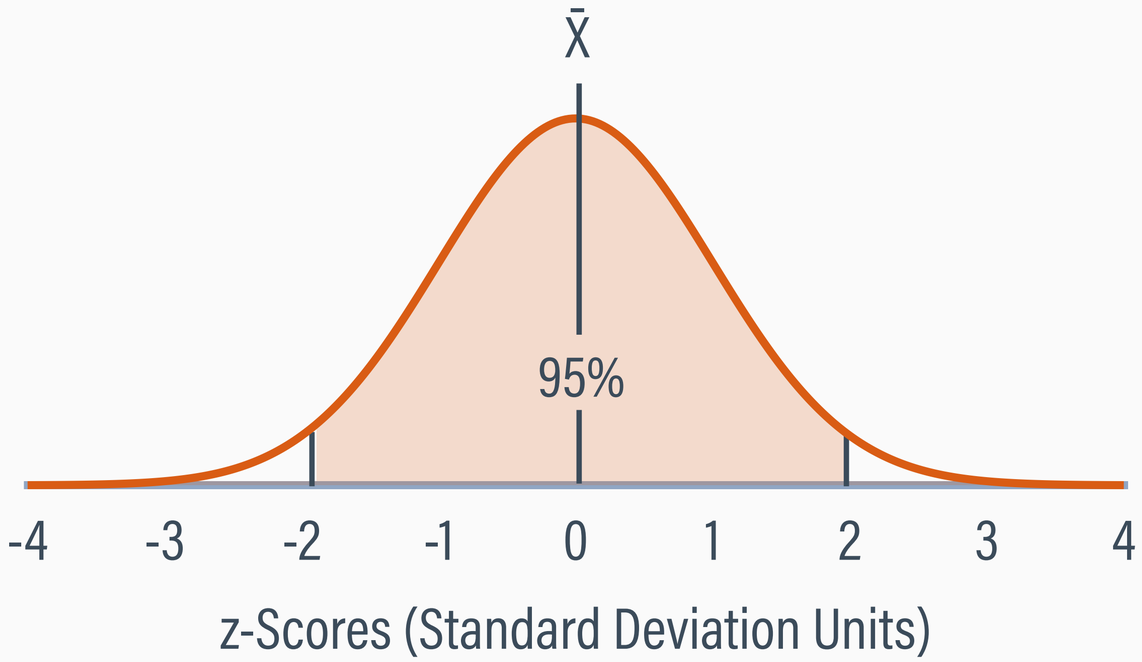
$$z = \frac{X - \bar{X}}{s} = \frac{4 - 10}{7.16} = \frac{-6}{7.16} = -0.84$$

X	z
19	1.26
19	1.26
17	0.98
14	0.56
7	−0.42
5	−0.7
4	−0.84
3	−0.98
2	−1.12

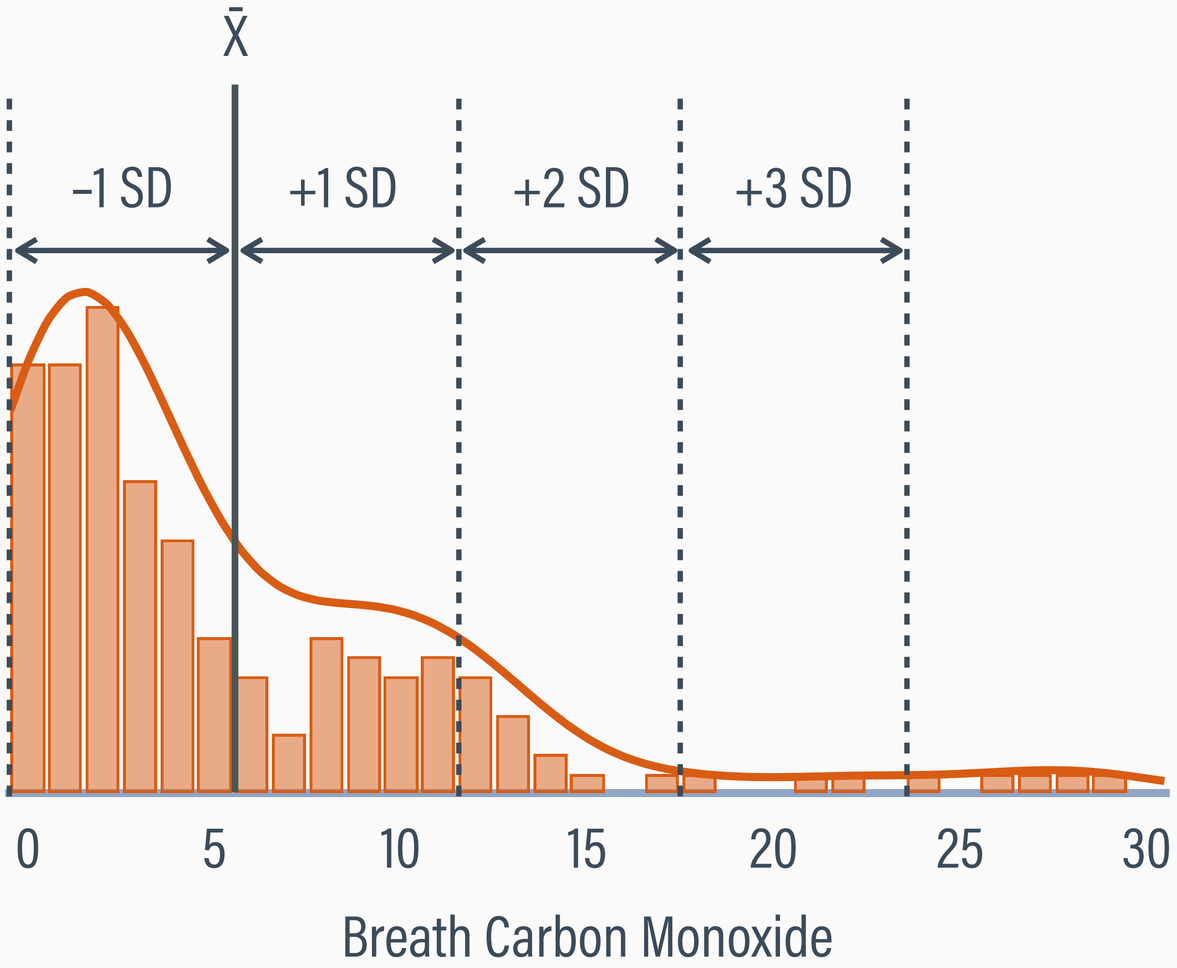
RULE OF THUMB FOR NORMAL DATA

- In a normal curve, 68% of the scores are within ± 1 z-scores (standard deviation units) of the mean, and 95% are within ≈ ± 2





CLINICAL TRIAL DESCRIPTIVES REVISITED

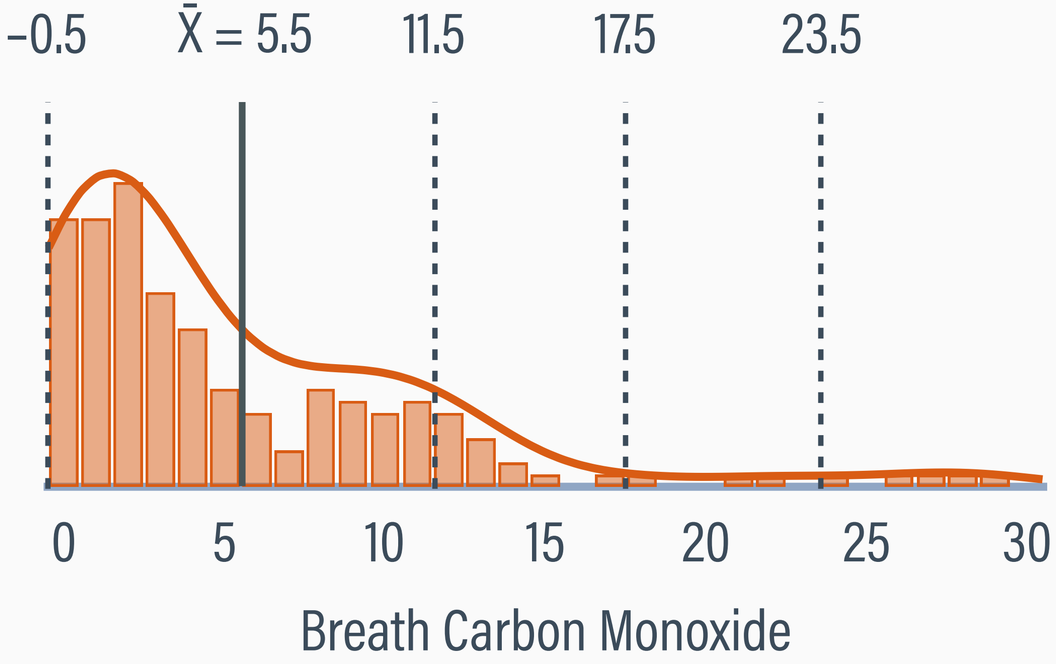


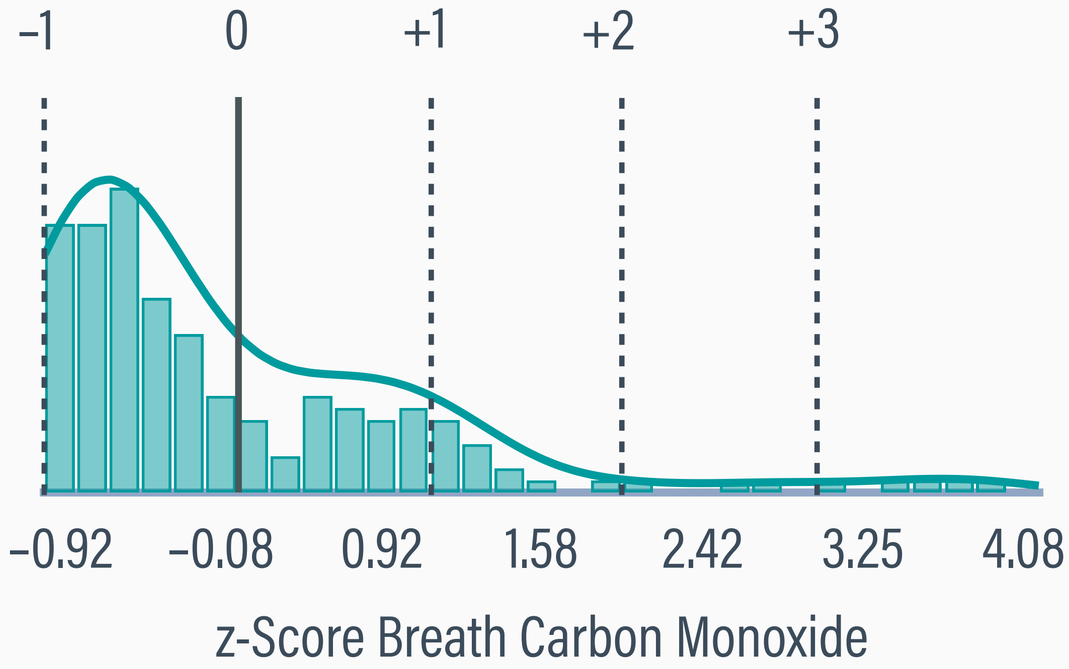
Mean (X̄)	5.5
Std. Dev. (s)	6
N	165

z-SCORES AND DISTRIBUTION SHAPE

- The z-score transformation changes the mean to 0 and standard deviation to 1, but it does not change the distribution's shape

Called 'linear transformation'

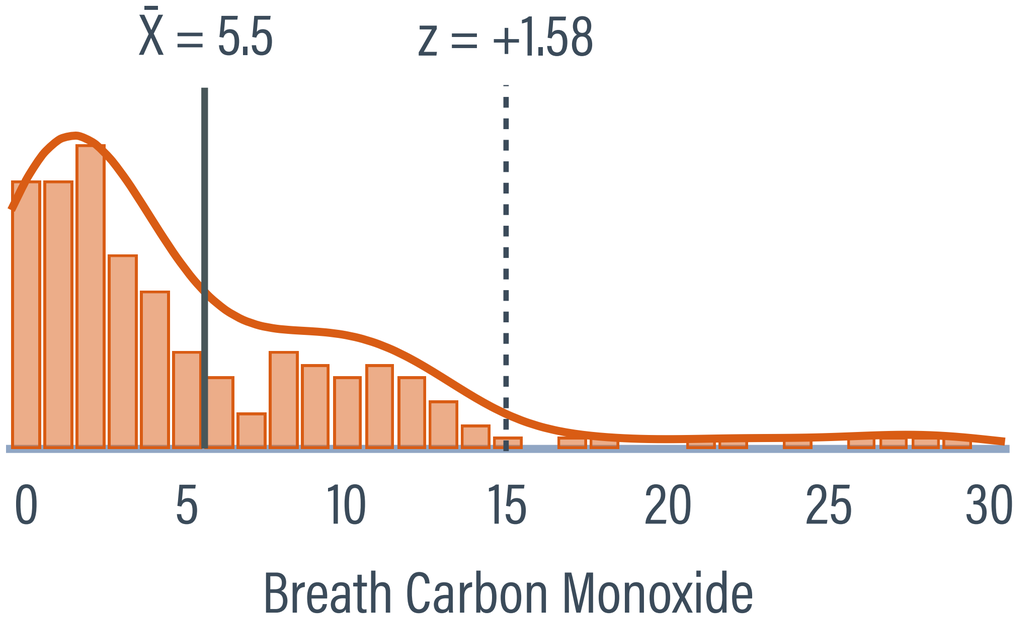




R OUTPUT

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
COWeek8	1	165	5.53	5.96	3.00	4.55	2.97	0.00	29.00	29.00	1.75	3.35	0.46
zCOWeek8	2	165	0.00	1.00	-0.42	-0.16	0.50	-0.93	3.94	4.87	1.75	3.35	0.08





The mean and standard deviation are $\bar{X} = 5.5$ and $s = 6$. The z-score for someone with a breath CO value of 15 is about $z = 1.58$. In small groups of two or three, interpret the z-score in practical terms. Your explanation should reference both the sign and the magnitude of the value. How extreme is this score?

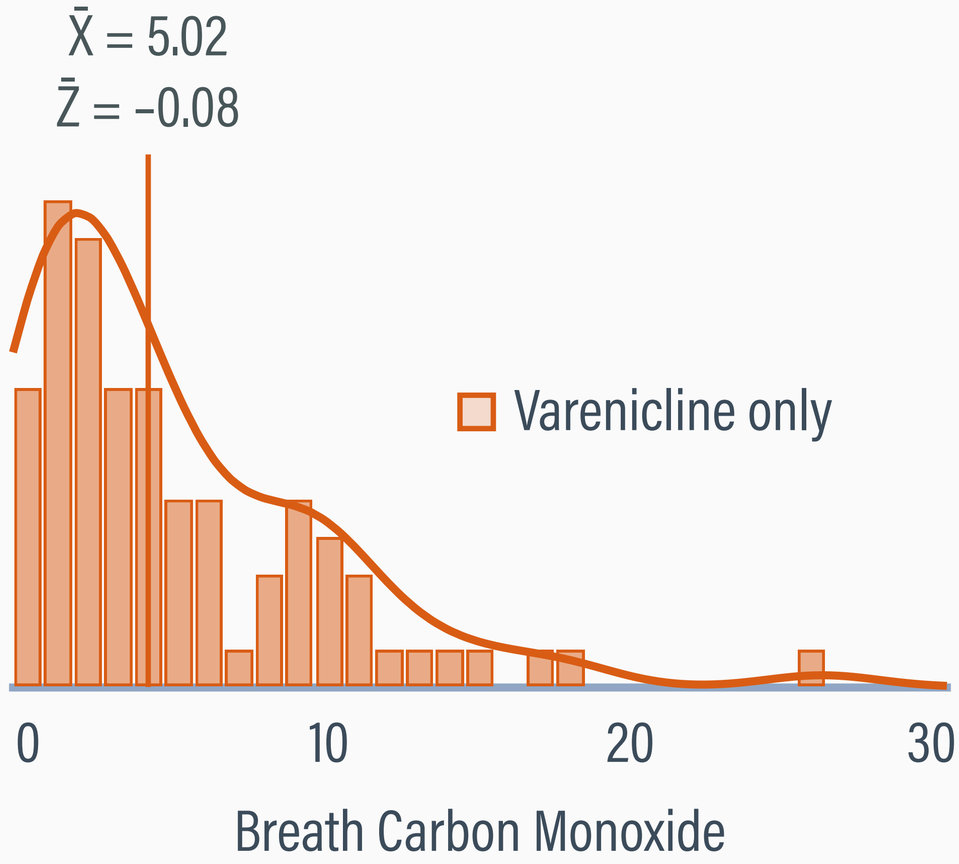
COMPARATIVE RESEARCH QUESTIONS

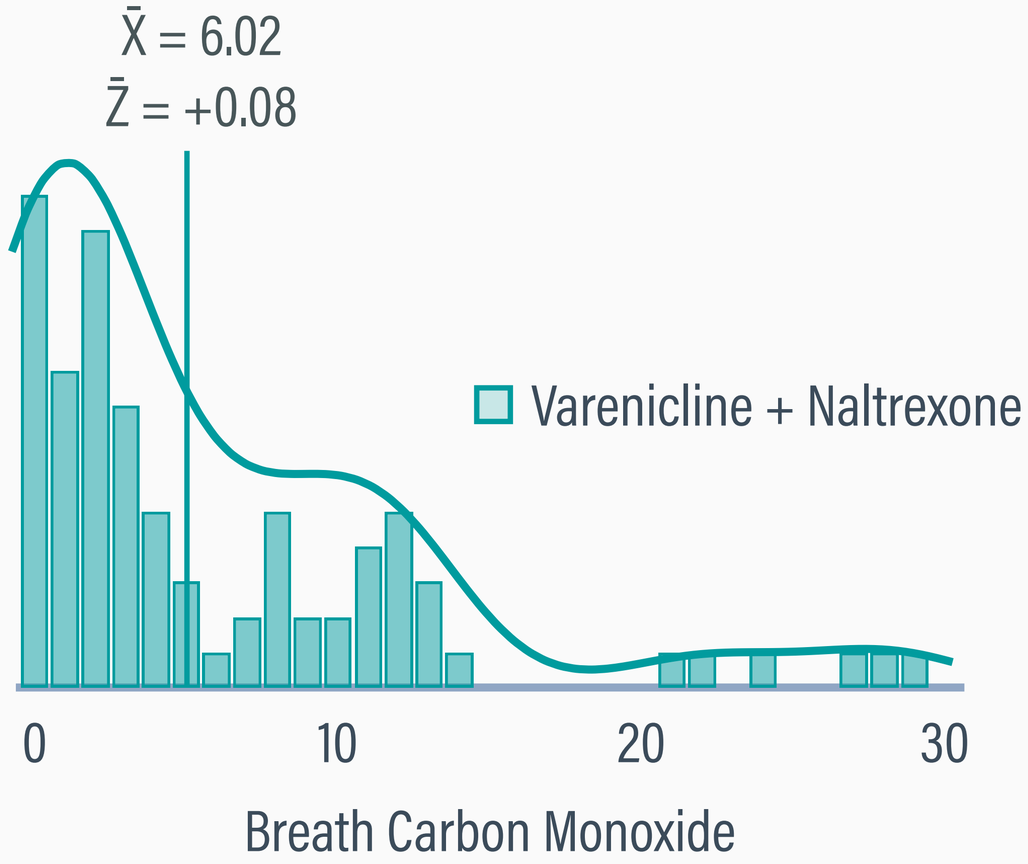
- Comparative research questions ask whether two or more groups (or occasions) differ from one another
- Question: Do participants in the two treatment groups differ in their smoking levels?
- We can answer this question by comparing descriptive statistics

STANDARDIZED MEAN DIFFERENCE PREVIEW

- It is often difficult to assess the magnitude of a group mean difference on the raw metric of the data
- A common approach is to express the mean difference in standard deviation (z-score) units
- The mean difference in z-score units is called the standardized mean difference effect size

GROUP MEANS AS z-SCORES



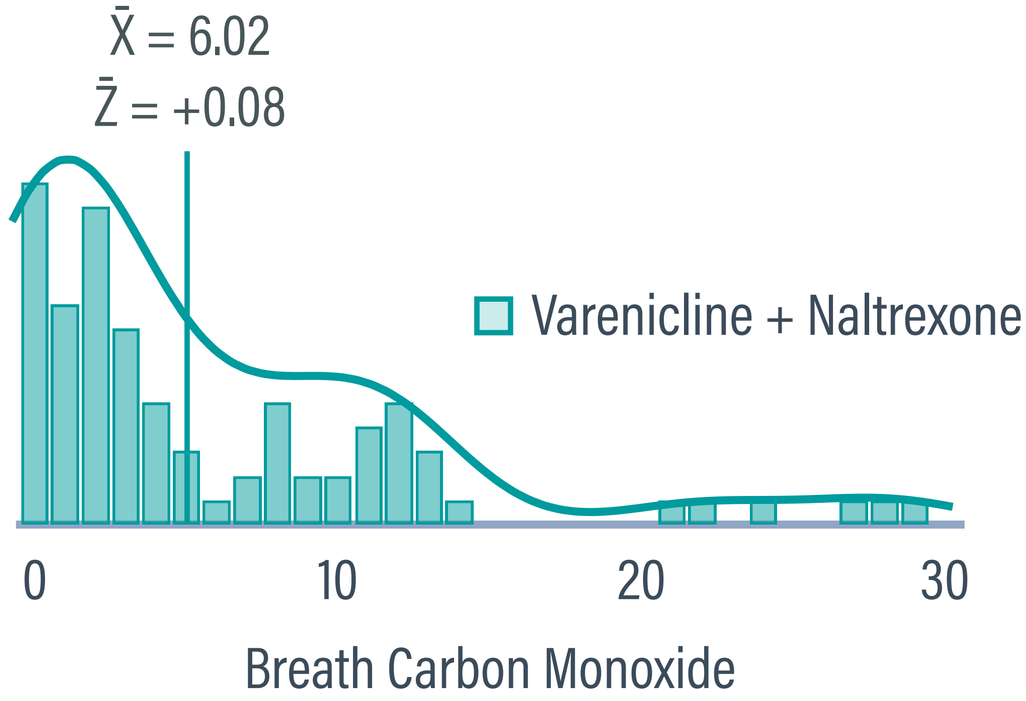


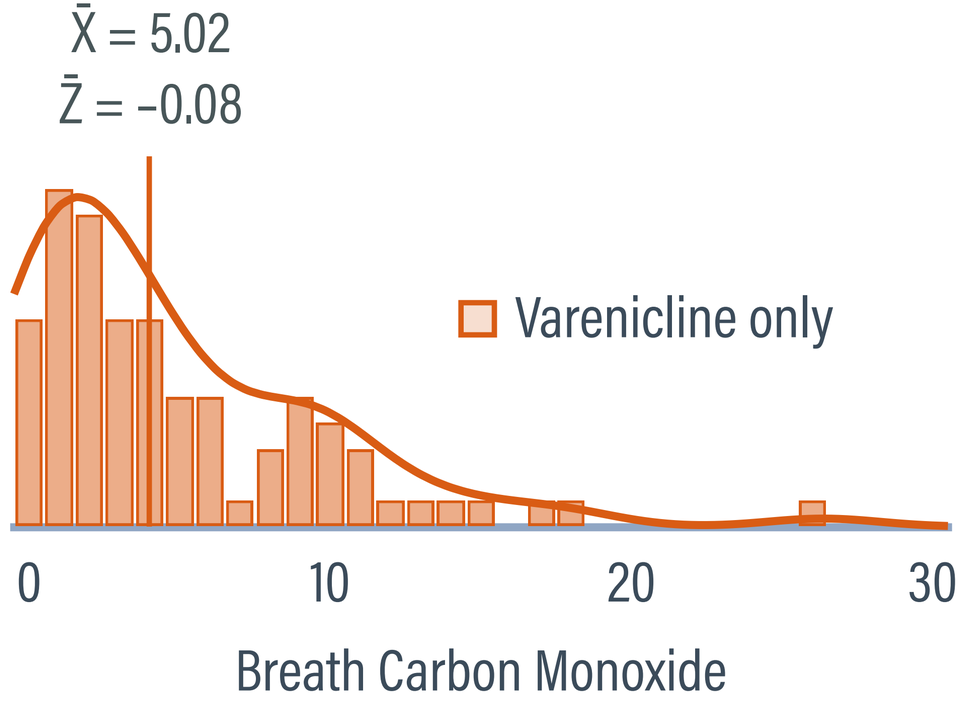
R OUTPUT

```
 Descriptive statistics by group
Condition: Varenicline
          vars  n  mean   sd median trimmed  mad   min  max range skew kurtosis   se
zCOWeek8     1 82 -0.08 0.82  -0.34    -0.2 0.62 -0.93 3.44  4.36 1.57     3.09 0.09
------------------------------------------------------------------------------------
Condition: Varenicline + Naltrexone
          vars  n mean   sd median trimmed  mad   min  max range skew kurtosis   se
zCOWeek8     1 83 0.08 1.15  -0.42   -0.12 0.75 -0.93 3.94  4.87 1.62     2.31 0.13
```



One of the main goals of the research study is to determine whether the treatment groups differ. In small groups of two or three, discuss the size of the mean difference on the z-score metric (about 0.16 standard deviation units). How would you gauge the magnitude of the standardized difference? Is it meaningful? This is a subjective evaluation, so provide a rationale.





OUTLINE

1. Measures of spread: Range, standard deviation, and variance
2. Statistical properties of the variance and standard deviation
3. Psychology application: Smoking cessation clinical trial
4. Standardizing: Expressing scores in standard deviation units
5. Study questions

STUDY QUESTIONS

- The study questions continue using the clinical trial research scenario introduced in the lecture, including new variables
- The questions illustrate the kinds of statistical reasoning the exams will require, including interpreting evidence, evaluating conclusions, and connecting graphical and numerical results
- Midterm exam questions will require you to apply these same reasoning skills to new research scenarios

R OUTPUT

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Participant	1	165	83.00	47.78	83	83.00	60.79	1	165	164	0.00	-1.22	3.72
Condition*	2	165	1.50	0.50	2	1.50	0.00	1	2	1	-0.01	-2.01	0.04
Male*	3	165	1.61	0.49	2	1.63	0.00	1	2	1	-0.43	-1.83	0.04
COWeek0	4	165	10.55	6.86	9	9.82	5.93	0	46	46	1.41	3.60	0.53
COWeek4	5	165	5.46	5.10	4	4.69	4.45	0	24	24	1.24	0.96	0.40
COWeek8	6	165	5.53	5.96	3	4.55	2.97	0	29	29	1.75	3.35	0.46
QuitCigsWeek4*	7	165	1.62	0.49	2	1.65	0.00	1	2	1	-0.48	-1.78	0.04
QuitCigsWeek8*	8	165	1.64	0.48	2	1.68	0.00	1	2	1	-0.59	-1.66	0.04
DrinksWeek0	9	165	6.40	4.42	6	5.77	2.97	1	35	34	2.44	10.46	0.34
DrinksWeek4	10	165	3.59	2.98	3	3.30	2.97	0	13	13	0.80	0.30	0.23
DrinksWeek8	11	165	3.23	2.68	3	2.97	2.97	0	13	13	0.89	0.67	0.21
CigsWeek0	12	165	14.22	8.22	12	13.05	5.93	3	51	48	1.59	3.20	0.64
CigsWeek4	13	165	4.18	5.70	2	3.08	2.97	0	41	41	3.01	13.13	0.44
CigsWeek8	14	165	3.16	4.75	2	2.25	2.97	0	35	35	3.44	16.81	0.37

STUDY QUESTIONS (1)

1. One of the key variables is the number of cigarettes smoked during the last week (CigsWeek8). At the 8-week follow-up, scores ranged between 0 and 35. Make up a sample of five scores that exhibit substantial variability.

2. Make up a sample of five smoking scores that exhibit very little variability.

3. Make up a sample of five smoking scores with a standard deviation equal to zero.

STUDY QUESTIONS (2)

4. Still considering the number of cigarettes smoked during the last week (CigsWeek8), the mean is 3.16. What is a distance or deviation score in this context? Illustrate a deviation score for an individual with a high smoking score. Do the same for a low smoking score.

5. Why are deviation scores squared when computing the variance and standard deviation? What problem would occur if the raw deviation scores were simply averaged?

STUDY QUESTIONS (3)

6. The maximum value of CigsWeek0 is 51. Suppose this score were changed from 51 to 14 (close to the mean). What would likely happen to the numeric value of the standard deviation and why? Answer needed

7. The sample variance (and standard deviation) is computed by dividing by N in the denominator is biased. Describe the concept of bias in this context. Answer needed

8. The benefit of computing the standard deviation by dividing by N – 1 rather than N depends on the sample size. Does you think this choice would make a material difference in this example?

STUDY QUESTIONS (4)

9. At the 8-week follow-up, the mean of CigsWeek8 is 3.16 cigarettes and its standard deviation is 4.75 cigarettes. Each of the following interpretations of the standard deviation is incorrect. For each statement, explain why it is incorrect and identify the statistical quantity or concept it describes. Then provide an accurate interpretation of the standard deviation of CigsWeek8.

   A. The difference between the highest and lowest smoking scores is 4.75 cigarettes.

   B. The average squared distance between the individual smoking scores and the mean is 4.75 cigarettes.

   C. After taking the absolute values of the deviation scores, the average absolute distance from the mean is 4.75 cigarettes.

STUDY QUESTIONS (5)

10. Breath carbon monoxide is measured in a physiological metric that may require specialized knowledge to interpret. To make scores easier to understand, suppose you convert them to z-scores. Provide a hypothetical z-score for someone whose breath carbon monoxide is substantially higher than the sample mean. Explain what the score's sign and magnitude mean.

11. Do the same for someone whose breath carbon monoxide is below the sample mean.

STUDY QUESTIONS (6)

12. Suppose the researchers want to determine whether the two medication conditions differ more in drinking or smoking at the 8-week follow-up. Explain why the raw differences between the condition means cannot be compared directly across the two outcomes. How could z-scores be used to answer this question? Answer needed

13. Suppose researchers want to examine the change in smoking between baseline and the 8-week follow-up. The raw means are CigsWeek0 = 14.22 and CigsWeek8 = 3.16. Suppose each variable were converted to z-scores separately. Could the two standardized variables be used to evaluate change over time? Explain why or why not. Hint: What happens to the means?